Lecture 6: Local Sequence Matching

		H	E	A	G	A	W	G	H	E	E
	0	0	0	0	0	0	0	0	0	0	0
P	0	0	0	0	**0**	0	0	0	0	0	0
A	0	0	0	5	0	**5**	0	0	0	0	0
W	0	0	0	0	2	0	**20**	**12**	4	0	0
H	0	10	2	0	0	0	12	18	**22**	14	6
E	0	2	16	8	0	0	4	10	18	**28**	20
A	0	0	8	21	13	5	0	4	10	20	27
E	0	0	6	13	18	12	4	0	4	16	26

$$F_{i,j} = \max \begin{cases} 0 \\ F_{i-1,j-1} + s(a_i, b_j) \\ F_{i-1,j} - 8 \\ F_{i,j-1} - 8 \end{cases}$$

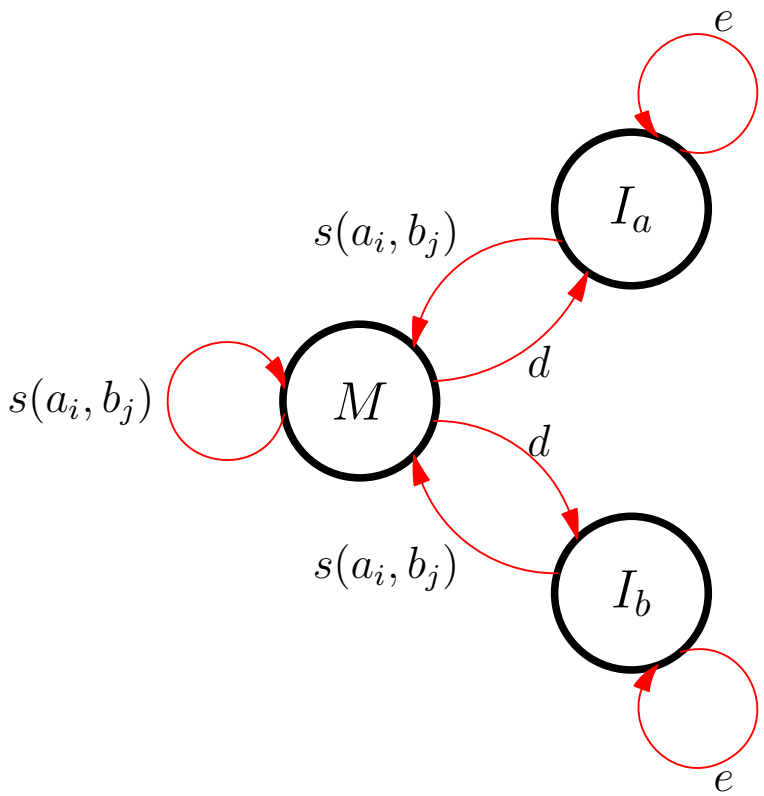
A W G H E

A W - H E

Local sequence matching, Gaps, FSA, Extreme Values

Outline

1. **Local Sequence Alignment**
2. Affine Gaps
3. BLAST and FASTA
4. Significance Scores



Local Alignment

- We saw how we could globally align sequences
- But the nature of evolution is that often sections from one part of the genome are accidentally copied to another
- Thus, we are often more interested in finding best matching sub-sequences
- In biology this problem is known as the Smith-Waterman problem after the authors that first posed the problem and gave an algorithm

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Dynamic Programming Solution

- Surprisingly, by a very small tweak to the global alignment problem, we can solve this problem too

- The trick is to add the option of ignoring the match with cost 0

$$F(i,j) = \max \begin{cases} 0 \\ F(i-1,j-1) + s(a_i, b_j) \\ F(i-1,j) + d \\ F(i,j-1) + d \end{cases}$$

(we use $d = -8$ below)

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Local Alignment

		H	E	A	G	A	W	G	H	E	E
P											
A											
W											
H											
E											
A											
E											

$$F_{i,j} = \max \begin{cases} 0 \\ F_{i-1,j-1} + s(a_i, b_j) \\ F_{i-1,j} - 8 \\ F_{i,j-1} - 8 \end{cases}$$

Local Alignment

		H	E	A	G	A	W	G	H	E	E
	0										
P											
A											
W											
H											
E											
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Local Alignment

		H	E	A	G	A	W	G	H	E	E
	0	0									
P											
A											
W											
H											
E											
A											
E											

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Local Alignment

		H	E	A	G	A	W	G	H	E	E
	0	0	0	0	0	0	0	0	0	0	0
P											
A											
W											
H											
E											
A											
E											

$$F_{i,j} = \max \begin{cases} 0 \\ F_{i-1,j-1} + s(a_i, b_j) \\ F_{i-1,j} - 8 \\ F_{i,j-1} - 8 \end{cases}$$

Local Alignment

		H	E	A	G	A	W	G	H	E	E
	0	0	0	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0	0	0	0
A											
W											
H											
E											
A											
E											

$$F_{i,j} = \max \begin{cases} 0 \\ F_{i-1,j-1} + s(a_i, b_j) \\ F_{i-1,j} - 8 \\ F_{i,j-1} - 8 \end{cases}$$

Local Alignment

		H	E	A	G	A	W	G	H	E	E
	0	0	0	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0	0	0	0
A	0										
W											
H											
E											
A											
E											

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Local Alignment

		H	E	A	G	A	W	G	H	E	E
	0	0	0	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0	0	0	0
A	0	0									
W											
H											
E											
A											
E											

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Local Alignment

		H	E	A	G	A	W	G	H	E	E
	0	0	0	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0	0	0	0
A	0	0	0								
W											
H											
E											
A											
E											

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Local Alignment

		H	E	A	G	A	W	G	H	E	E
	0	0	0	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0	0	0	0
A	0	0	0	5							
W											
H											
E											
A											
E											

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Local Alignment

		H	E	A	G	A	W	G	H	E	E
	0	0	0	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0	0	0	0
A	0	0	0	5	0						
W											
H											
E											
A											
E											

$$F_{i,j} = \max \begin{cases} 0 \\ F_{i-1,j-1} + s(a_i, b_j) \\ F_{i-1,j} - 8 \\ F_{i,j-1} - 8 \end{cases}$$

Local Alignment

		H	E	A	G	A	W	G	H	E	E
	0	0	0	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0	0	0	0
A	0	0	0	5	0	5					
W											
H											
E											
A											
E											

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Local Alignment

		H	E	A	G	A	W	G	H	E	E
	0	0	0	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0	0	0	0
A	0	0	0	5	0	5	0				
W											
H											
E											
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Local Alignment

		H	E	A	G	A	W	G	H	E	E
	0	0	0	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0	0	0	0
A	0	0	0	5	0	5	0	0	0	0	0
W											
H											
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Local Alignment

		H	E	A	G	A	W	G	H	E	E
	0	0	0	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0	0	0	0
A	0	0	0	5	0	5	0	0	0	0	0
W	0										
H											
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Local Alignment

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	0	0	0	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0	0	0	0
A	0	0	0	5	0	5	0	0	0	0	0
W	0	0									
H											
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E											

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Local Alignment

		H	E	A	G	A	W	G	H	E	E
	0	0	0	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0	0	0	0
A	0	0	0	5	0	5	0	0	0	0	0
W	0	0	0								
H											
E											
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Local Alignment

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	0	0	0	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0	0	0	0
A	0	0	0	5	0	5	0	0	0	0	0
W	0	0	0	0							
H											
E											
A											
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		H	E	A	G	A	W	G	H	E	E
	0	0	0	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0	0	0	0
A	0	0	0	5	0	5	0	0	0	0	0
W	0	0	0	0	2						
H											
E											
A											
E											

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Local Alignment

		H	E	A	G	A	W	G	H	E	E
	0	0	0	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0	0	0	0
A	0	0	0	5	0	5	0	0	0	0	0
W	0	0	0	0	2	0					
H											
E											
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		H	E	A	G	A	W	G	H	E	E
	0	0	0	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0	0	0	0
A	0	0	0	5	0	5	0	0	0	0	0
W	0	0	0	0	2	0	20				
H											
E											
A											
E											

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	0	0	0	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0	0	0	0
A	0	0	0	5	0	5	0	0	0	0	0
W	0	0	0	0	2	0	20	12	4	0	0
H											
E											
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P	0	0	0	0	0	0	0	0	0	0	0
A	0	0	0	5	0	5	0	0	0	0	0
W	0	0	0	0	2	0	20	12	4	0	0
H	0	10	2	0	0	0	12	18	22	14	6
E											
A											
E											

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Local Alignment

		H	E	A	G	A	W	G	H	E	E
	0	0	0	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0	0	0	0
A	0	0	0	5	0	5	0	0	0	0	0
W	0	0	0	0	2	0	20	12	4	0	0
H	0	10	2	0	0	0	12	18	22	14	6
E	0	2	16	8	0	0	4	10	18	28	20
A	0	0	8	21	13	5	0	4	10	20	27
E	0	0	6	13	18	12	4	0	4	16	26

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		H	E	A	G	A	W	G	H	E	E
	0	0	0	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0	0	0	0
A	0	0	0	5	0	5	0	0	0	0	0
W	0	0	0	0	2	0	20	12	4	0	0
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P	0	0	0	0	0	0	0	0	0	0	0
A	0	0	0	5	0	5	0	0	0	0	0
W	0	0	0	0	2	0	20	12	4	0	0
H	0	10	2	0	0	0	12	18	**22**	14	6
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E

E

Local Alignment

		H	E	A	G	A	W	G	H	E	E
	0	0	0	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0	0	0	0
A	0	0	0	5	0	5	0	0	0	0	0
W	0	0	0	0	2	0	20	**12**	4	0	0
H	0	10	2	0	0	0	12	18	**22**	14	6
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H E

H E

Local Alignment

		H	E	A	G	A	W	G	H	E	E
	0	0	0	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0	0	0	0
A	0	0	0	5	0	5	0	0	0	0	0
W	0	0	0	0	2	0	**20**	**12**	4	0	0
H	0	10	2	0	0	0	12	18	**22**	14	6
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G H E

- H E

Local Alignment

		H	E	A	G	A	W	G	H	E	E
	0	0	0	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0	0	0	0
A	0	0	0	5	0	**5**	0	0	0	0	0
W	0	0	0	0	2	0	**20**	**12**	4	0	0
H	0	10	2	0	0	0	12	18	**22**	14	6
E	0	2	16	8	0	0	4	10	18	**28**	20
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W G H E

W - H E

Local Alignment

		H	E	A	G	A	W	G	H	E	E
	0	0	0	0	0	0	0	0	0	0	0
P	0	0	0	0	**0**	0	0	0	0	0	0
A	0	0	0	5	0	**5**	0	0	0	0	0
W	0	0	0	0	2	0	**20**	**12**	4	0	0
H	0	10	2	0	0	0	12	18	**22**	14	6
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A W G H E

A W - H E

Local Alignment

		H	E	A	G	A	W	G	H	E	E
	0	0	0	0	0	0	0	0	0	0	0
P	0	0	0	0	**0**	0	0	0	0	0	0
A	0	0	0	5	0	**5**	0	0	0	0	0
W	0	0	0	0	2	0	**20**	**12**	4	0	0
H	0	10	2	0	0	0	12	18	**22**	14	6
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A W G H E

A W - H E

Difference to Global Alignment

- The forward algorithm is the same except we insist $F(i, j) \geq 0$
- We also store the location of the highest $F(i, j)$
- In the backward algorithm we start at the best value and go back until we reach 0
- It is really very easy to change a global alignment to a local alignment

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Repeated and Overlapping Matches

- So far we have considered the single best match
- In same cases (e.g. where we have a repeated domain or motif) we may have one region being repeated many times within a second sequence
- We can modify our dynamic programming to allow this
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2. **Affine Gaps**
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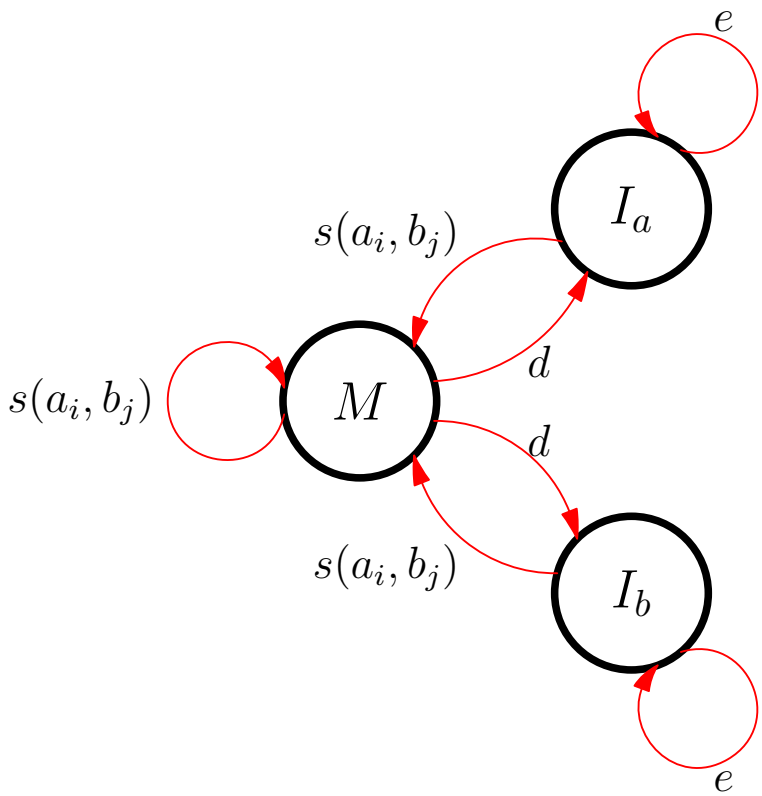
4. Significance Scores



Gaps

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- Deletions would result from jumping a number of bases or treating an exon as an intron
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Dumb Gap Algorithm

- If we assign a cost $\gamma(g)$ to a gap of size g then we could use the dynamic programme

$$F(i,j) = \max \begin{cases} F(i-1,j-1) + s(a_i, b_j), & \\ \quad F(k,j) - \gamma(i-k) & k = 0, \ldots, i-1 \\ \quad F(i,k) - \gamma(j-k) & k = 0, \ldots, j-1 \end{cases}$$

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Affine Gap

- If we choose a gap cost of

$$\gamma(g) = d + (g - 1)\, e$$

  then there exists a $O(n^2)$ algorithm to do matching

- An affine transform of a variable x is of the form $a\,x + b$ (i.e. a straight line)—note that $\gamma(g)$ is an affine function

- Using $d = -12$ and $e = -2$ gives a good model

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- To efficiently cope with gaps we need to introduce three matrices (rather than just $F(i,j)$)
  - ⋆ $M(i,j)$ is the best score up to (i,j) given a_i and b_j align
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Update Equation

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$$I_a(i,j) = \max \begin{cases} M(i-1,j) - d \\ I_a(i-1,j) - e \end{cases}$$

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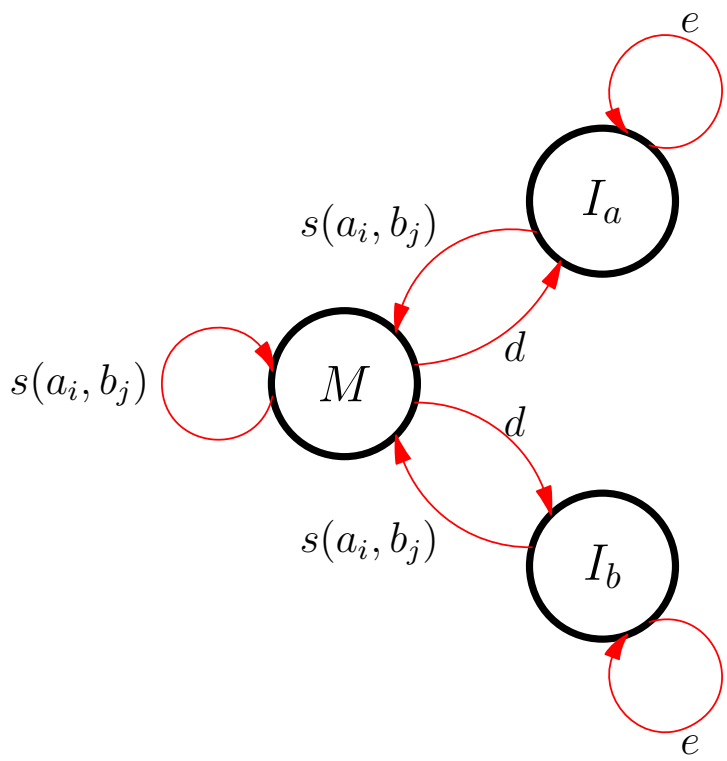
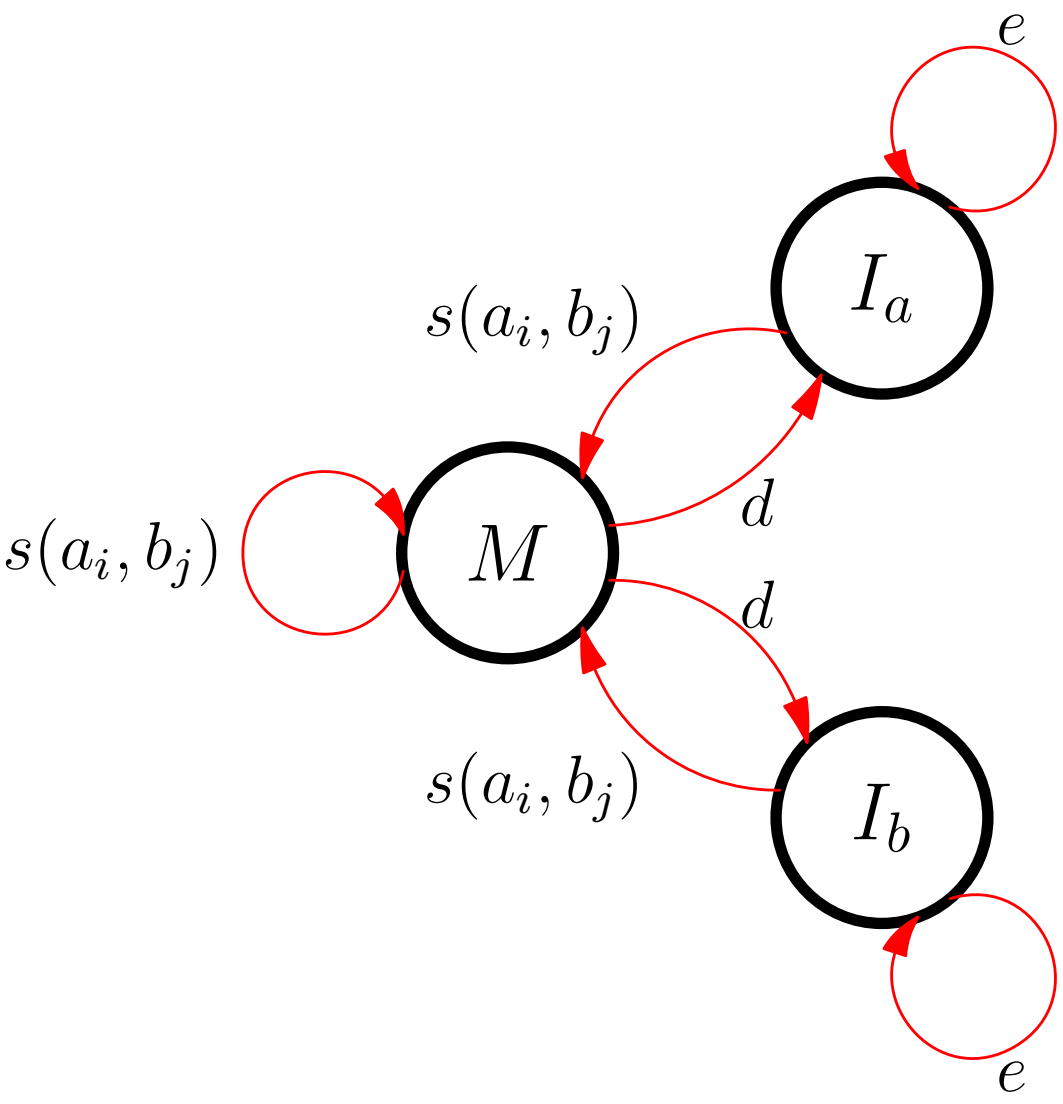


Finite State Automaton

Matching with Gaps



V L S P A D - K

H L - - A E S K

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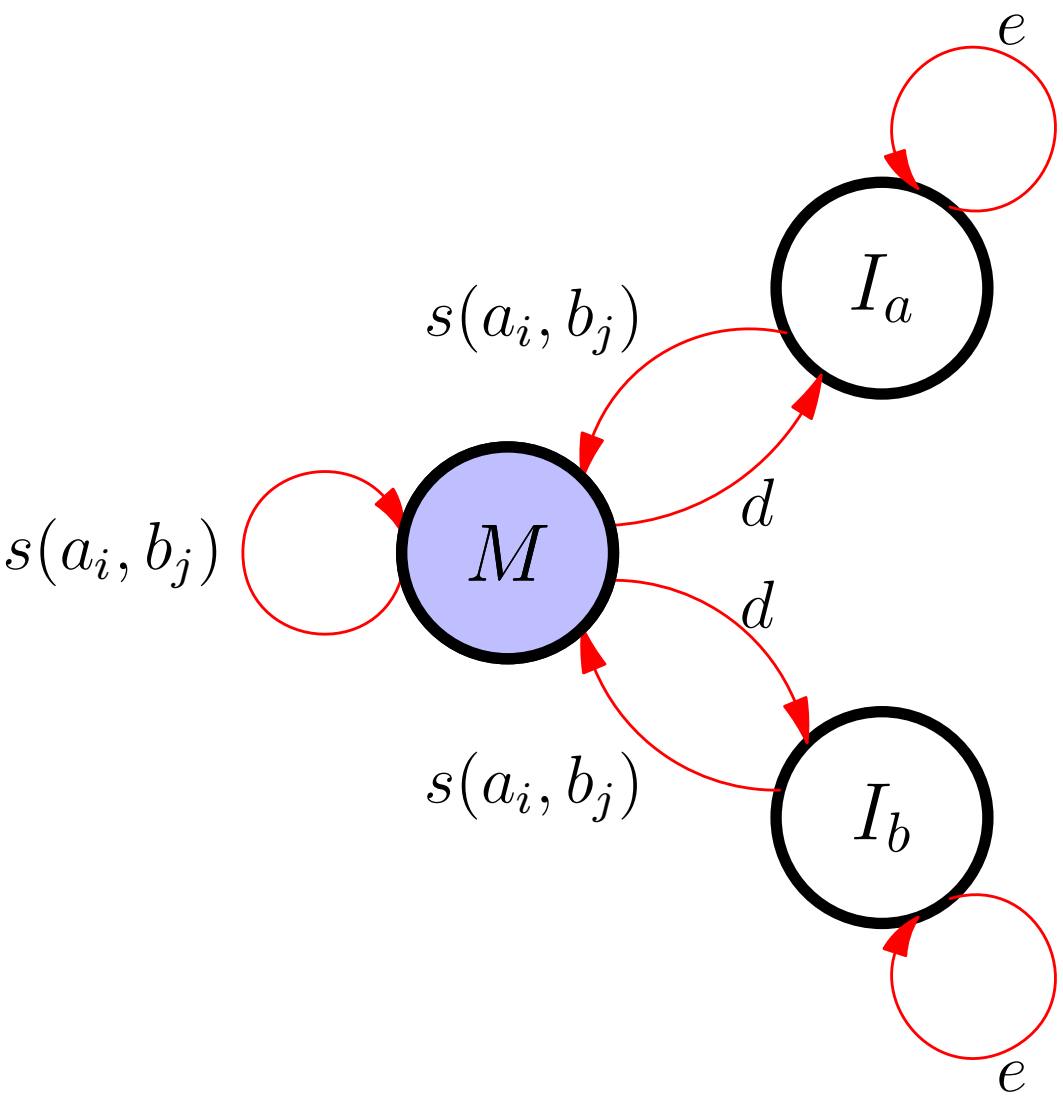

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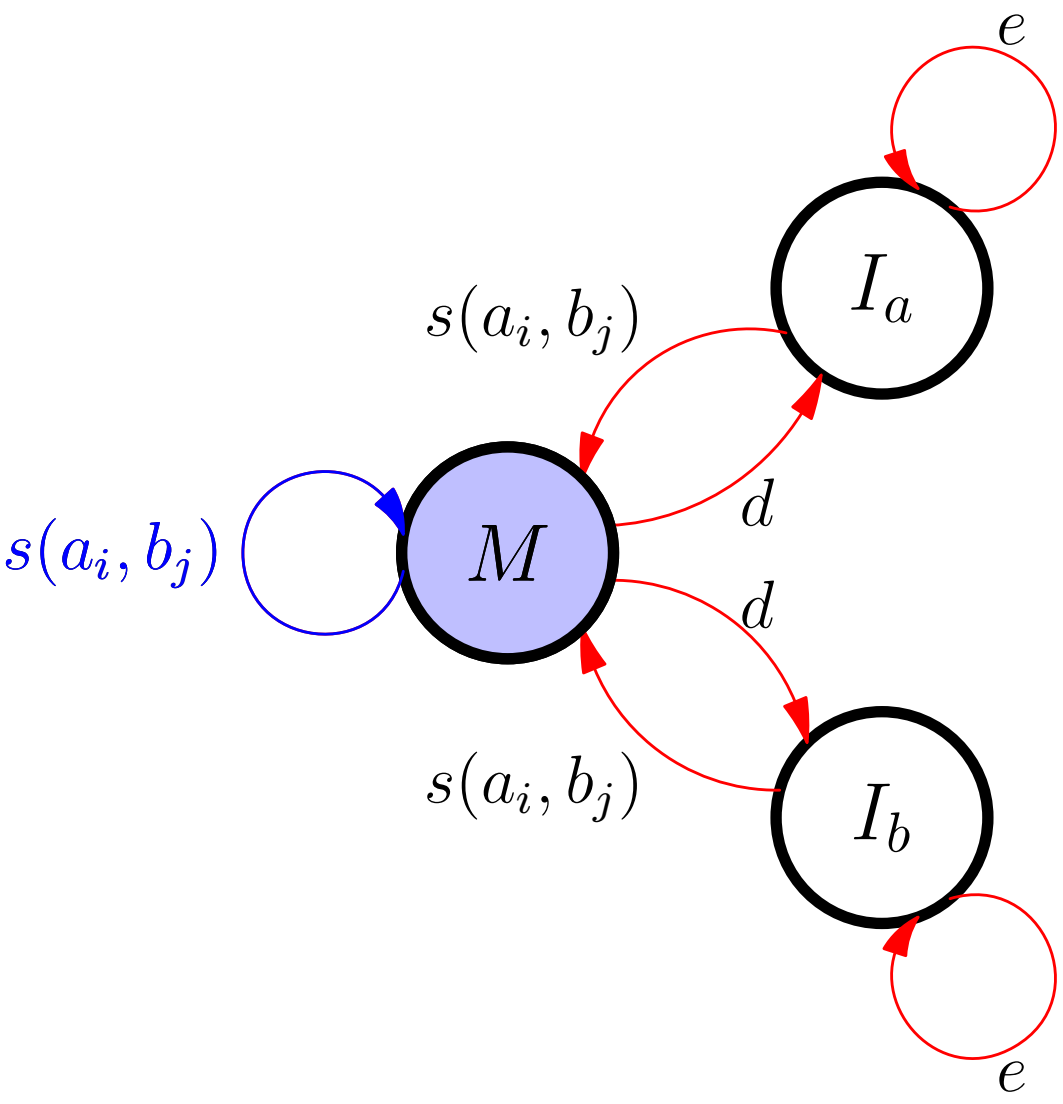

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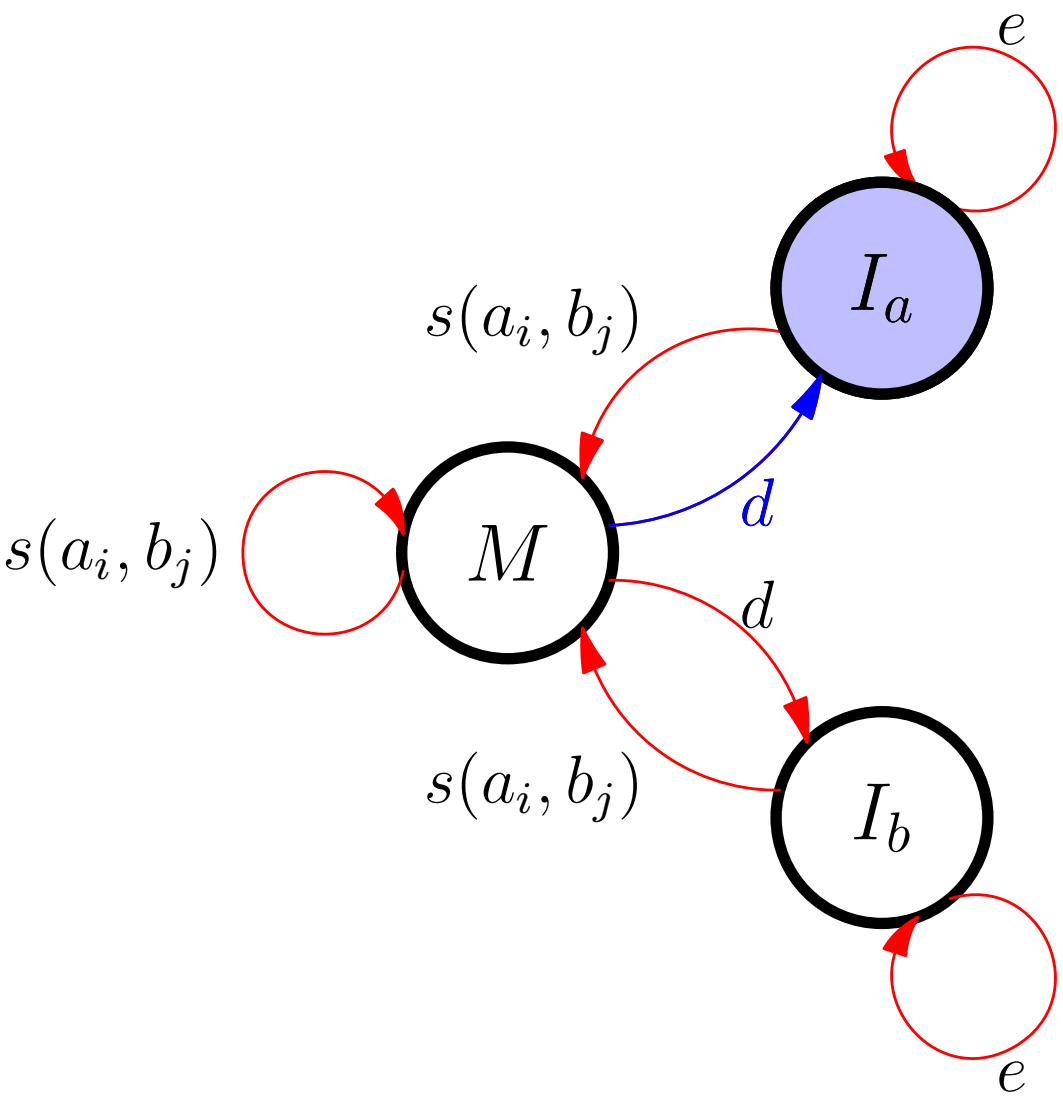

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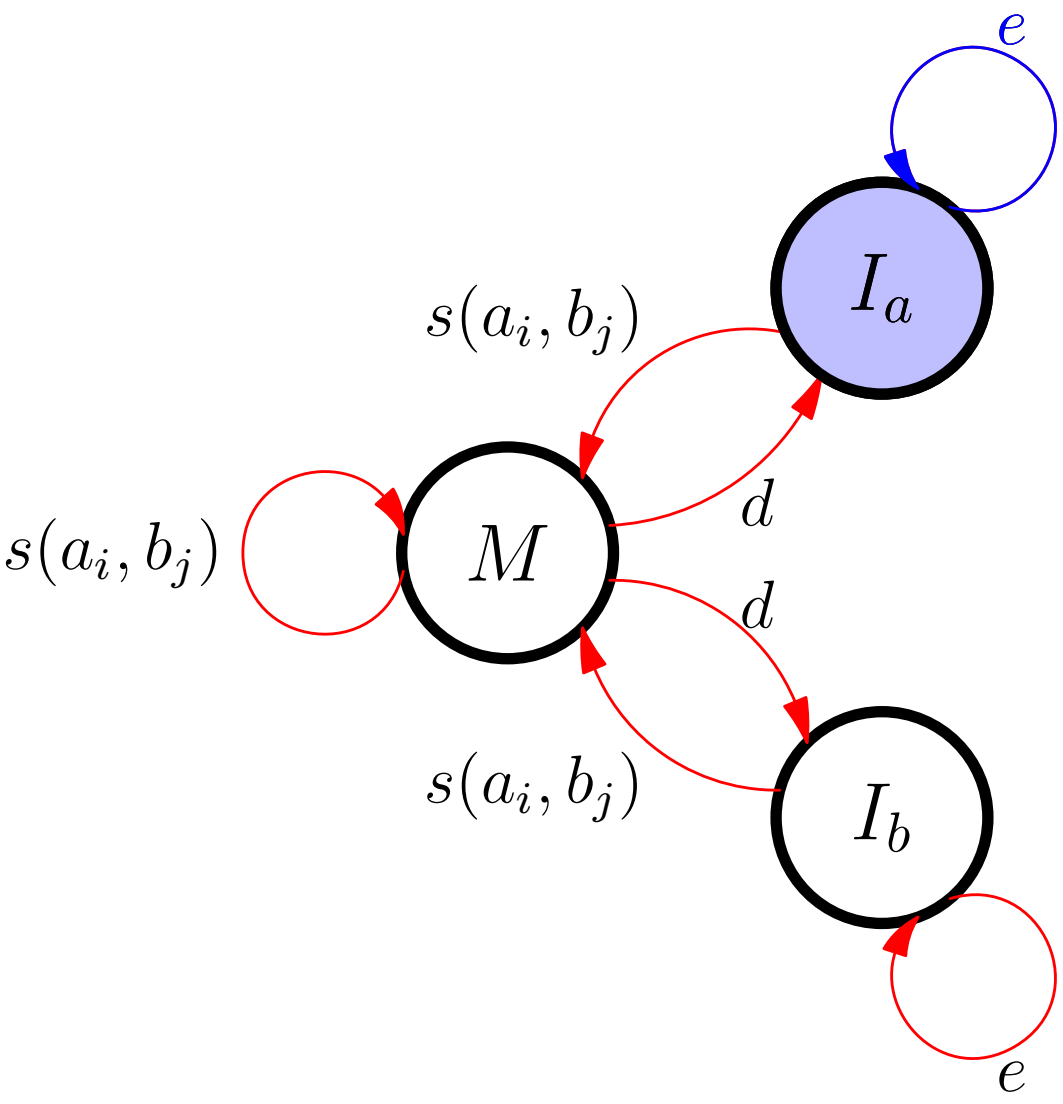

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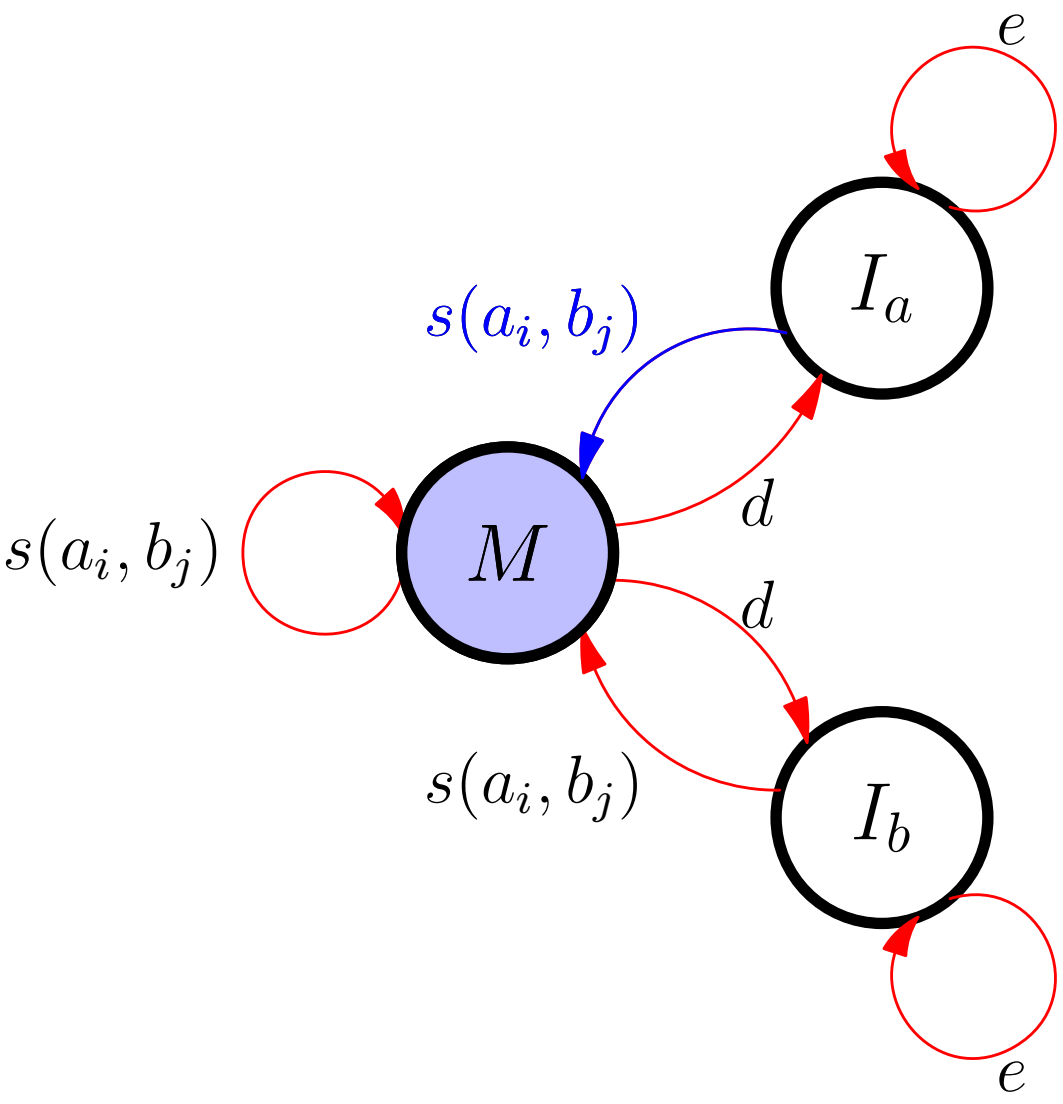

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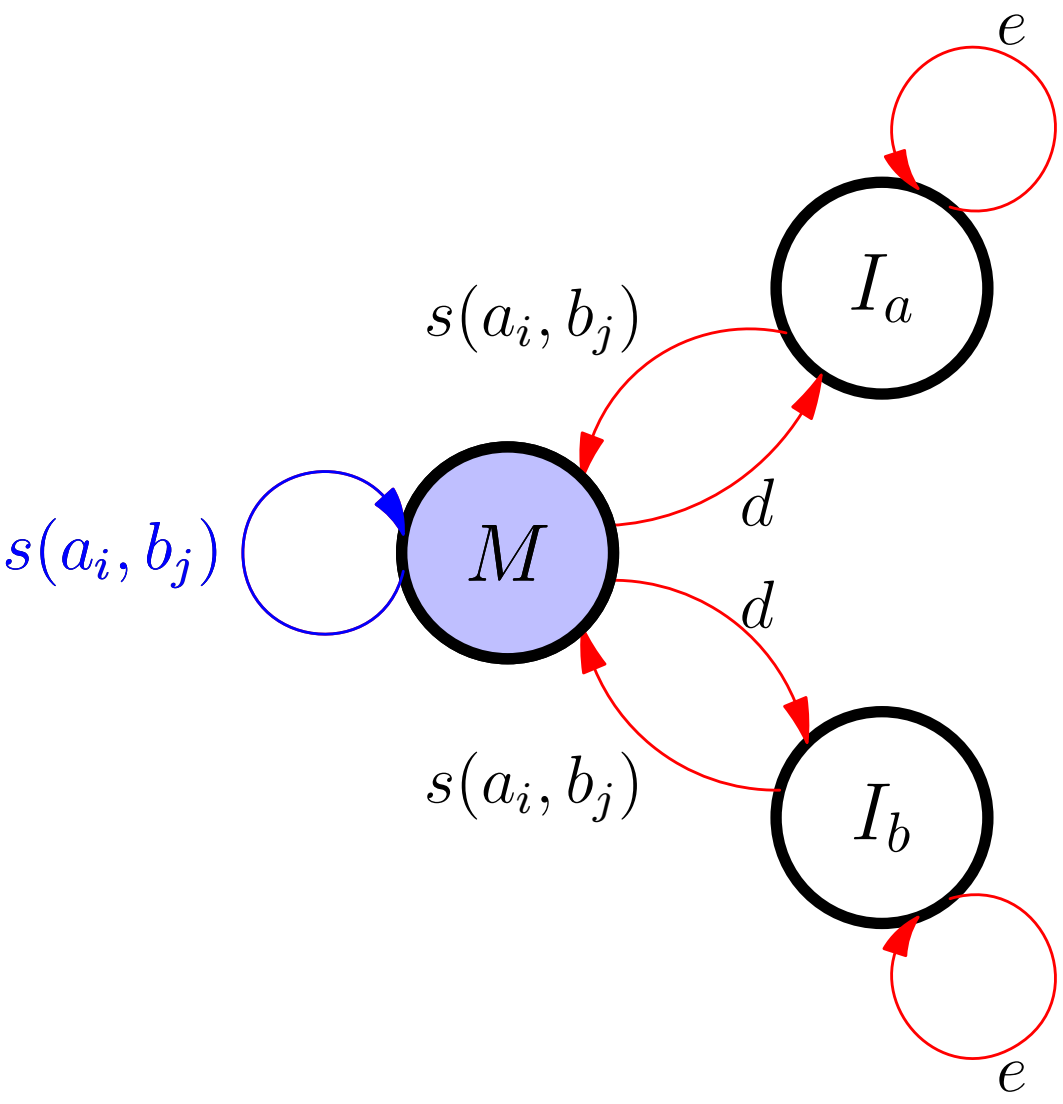

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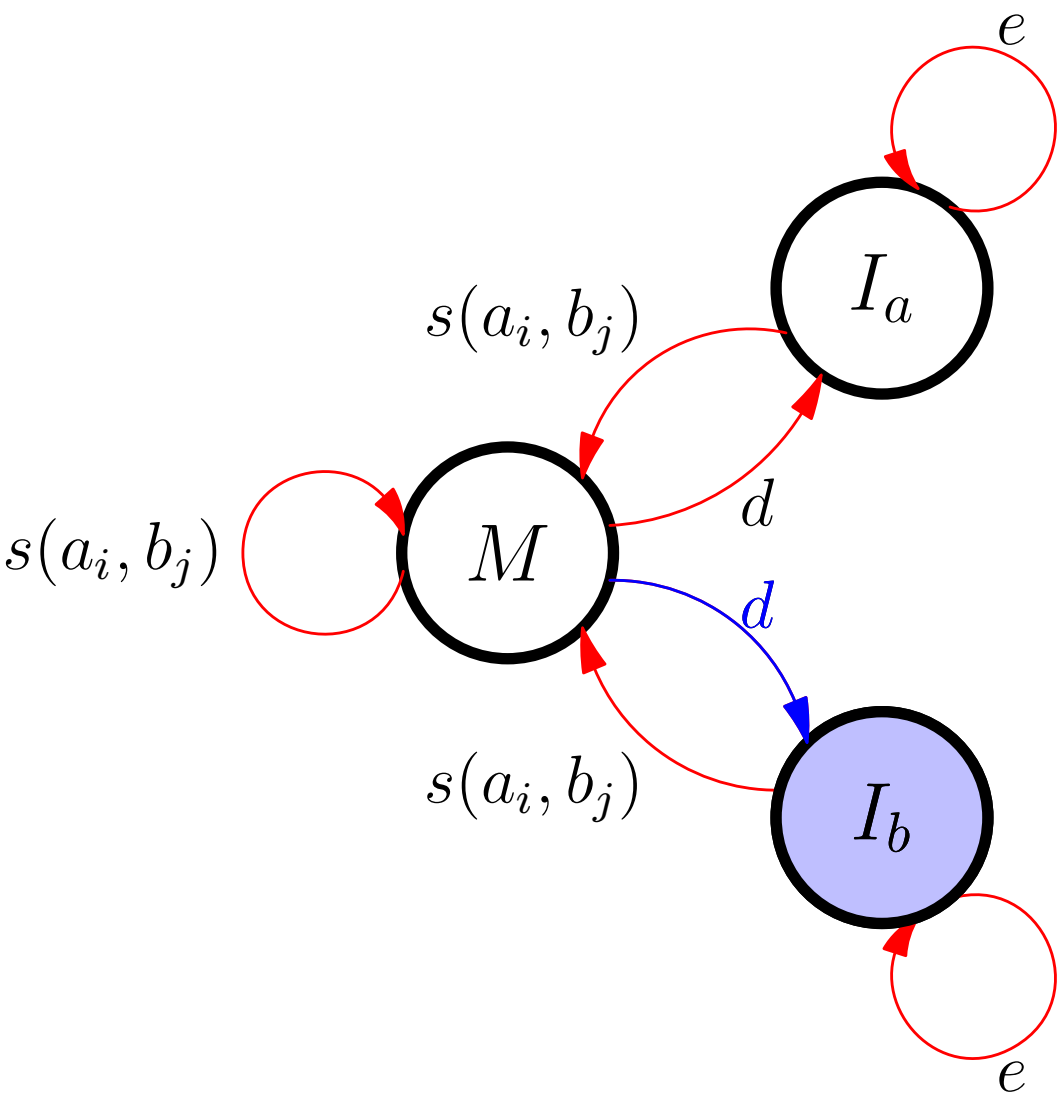

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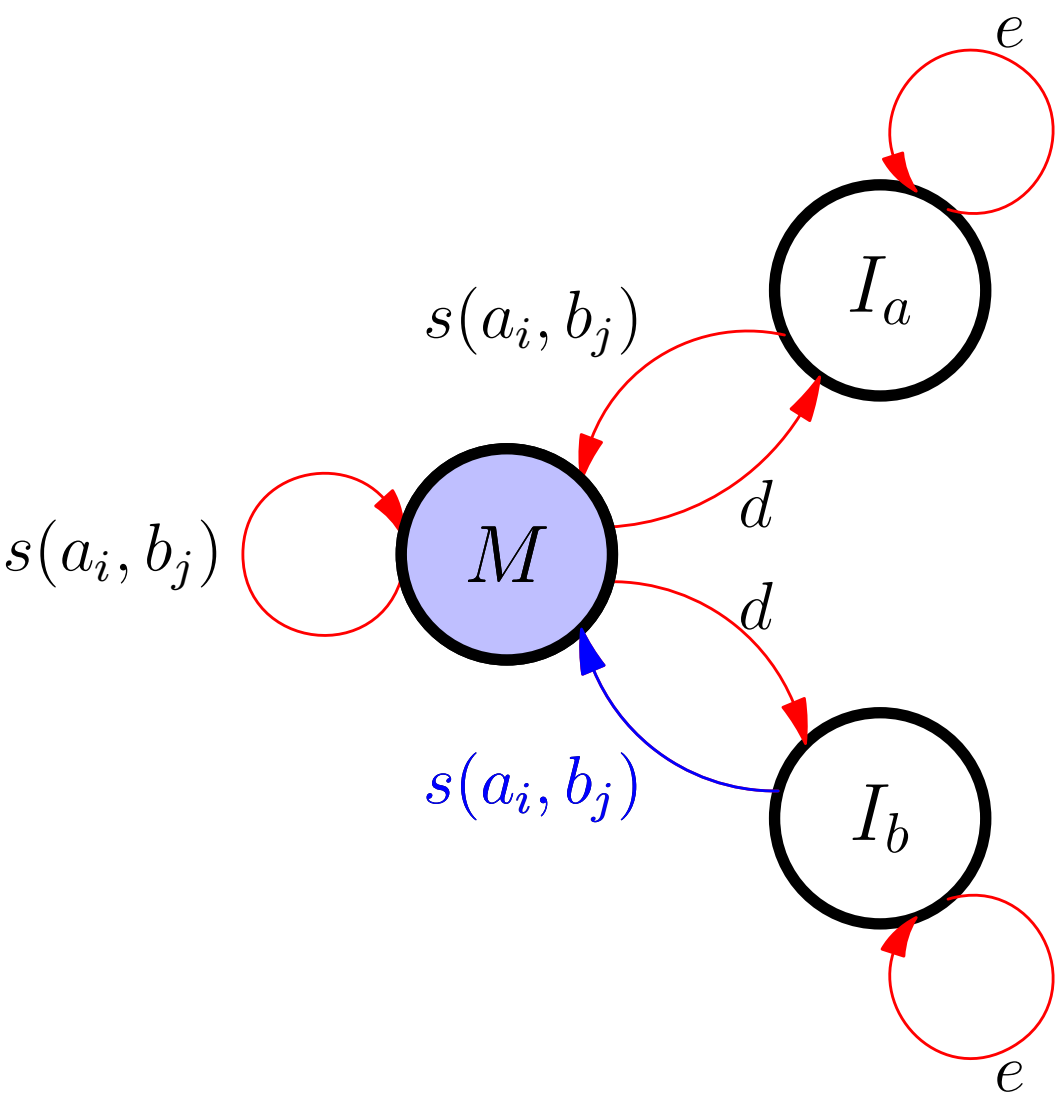

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- The finite state automaton (FSA) description provides an elegant visualisation of dynamic programming rules
- It makes it much easier to come up with different models that perhaps can better capture the evolutionary process
- But it comes at a cost that for each new state you need to generate an extra matrix—slowing down the matching
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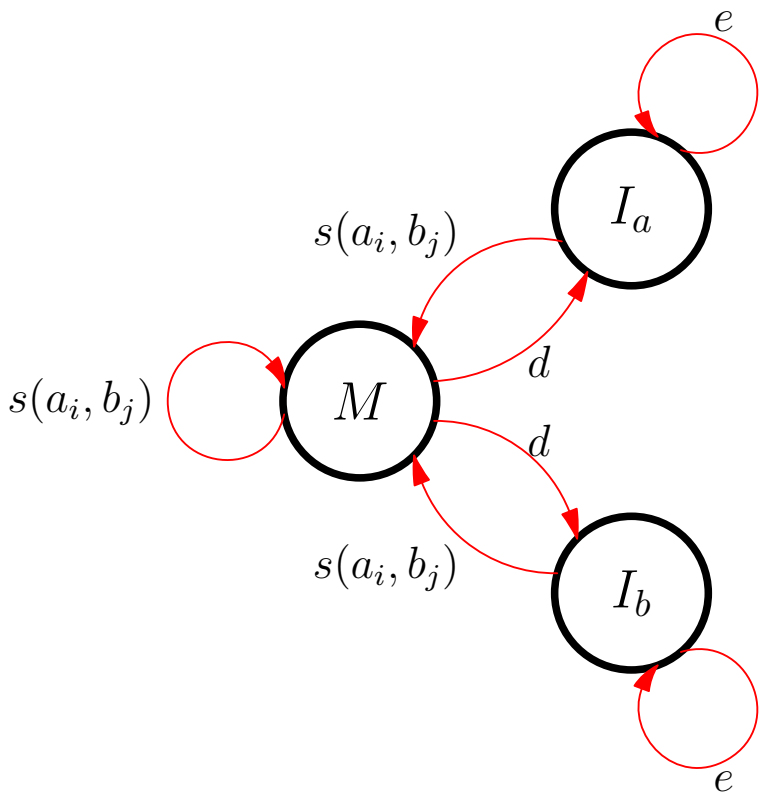

Inexact Matching in Practice

- Proteins are a few 100 amino acids long so $O(mn)$ is doable
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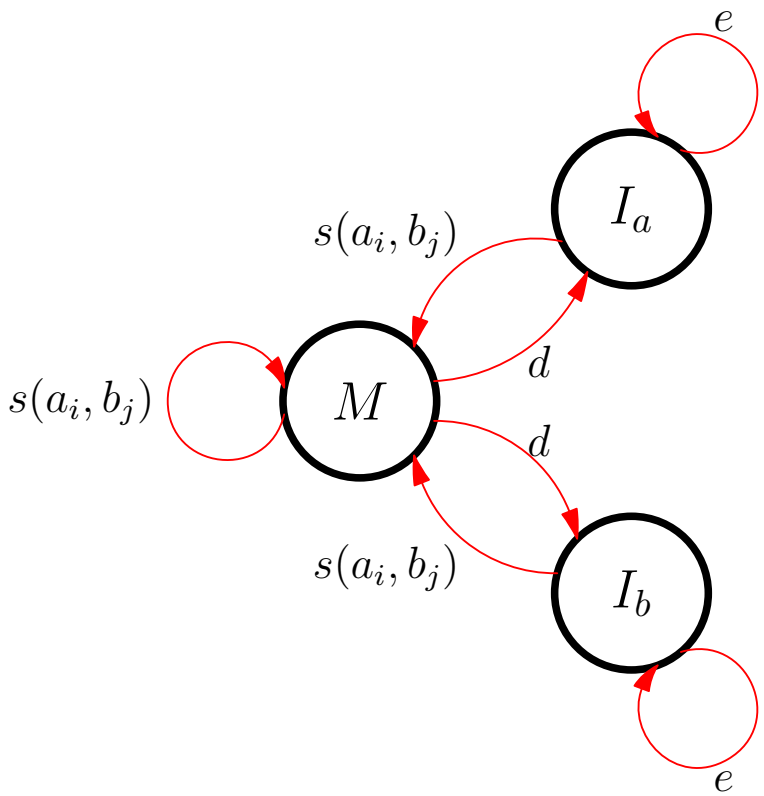

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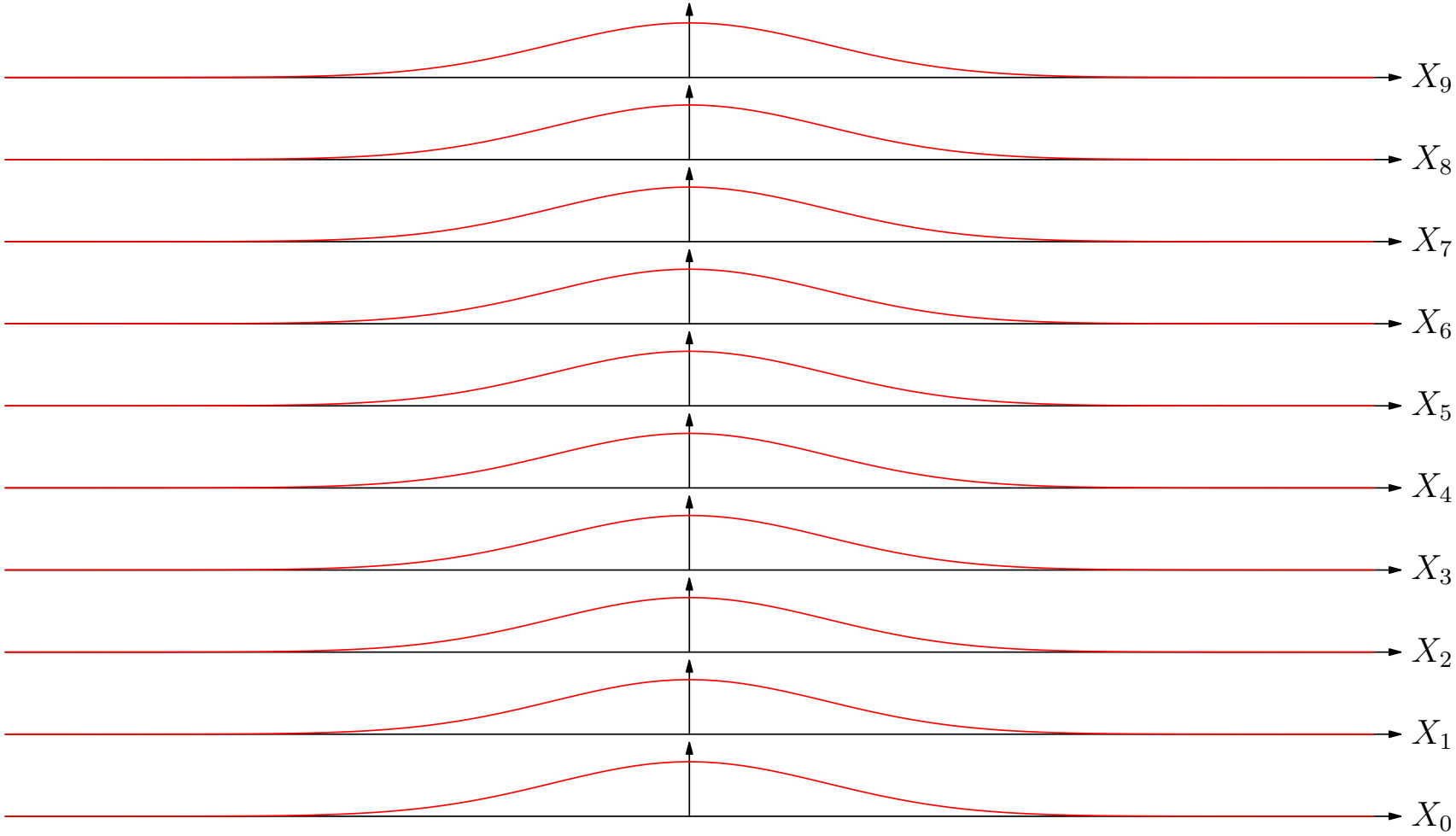

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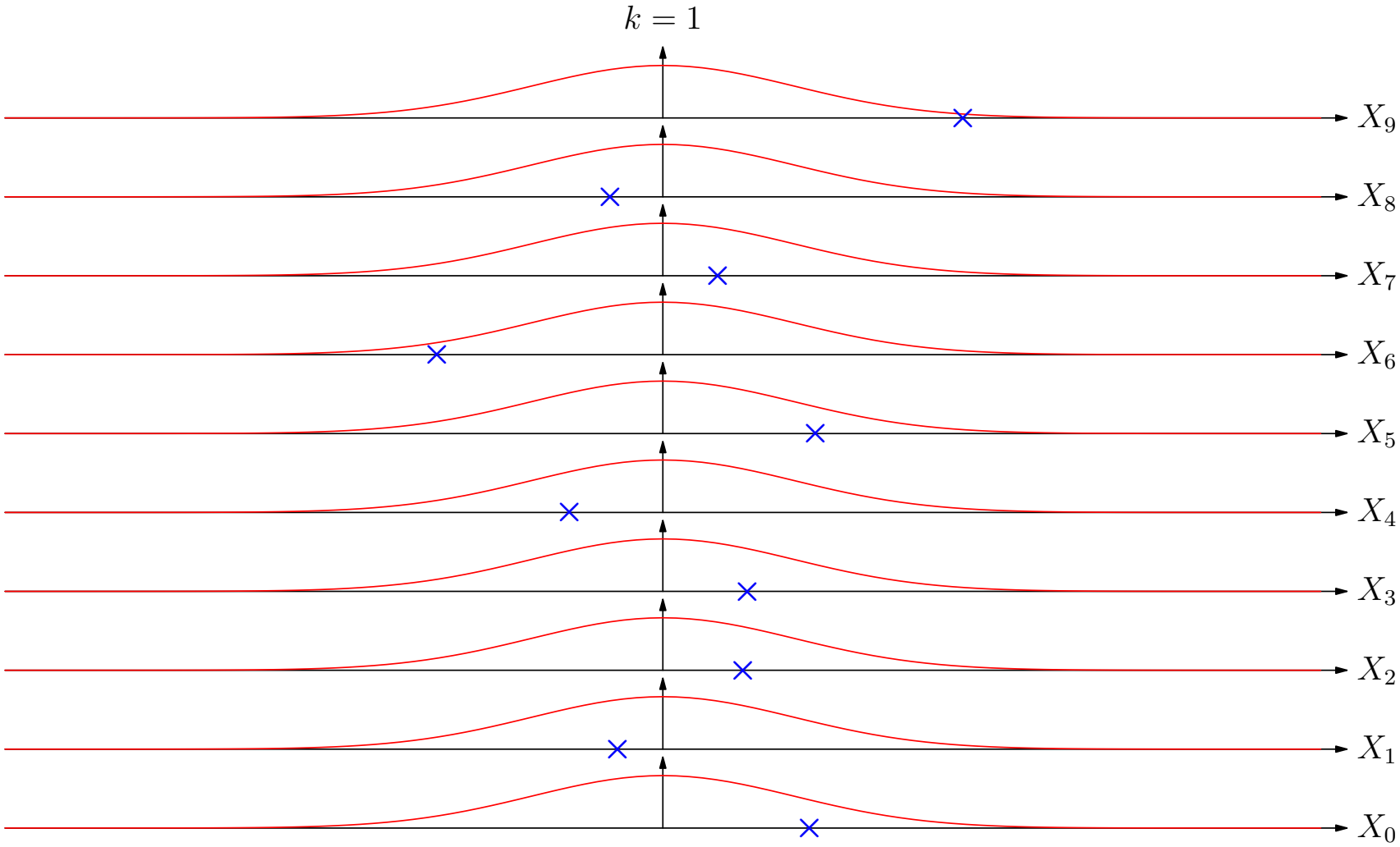

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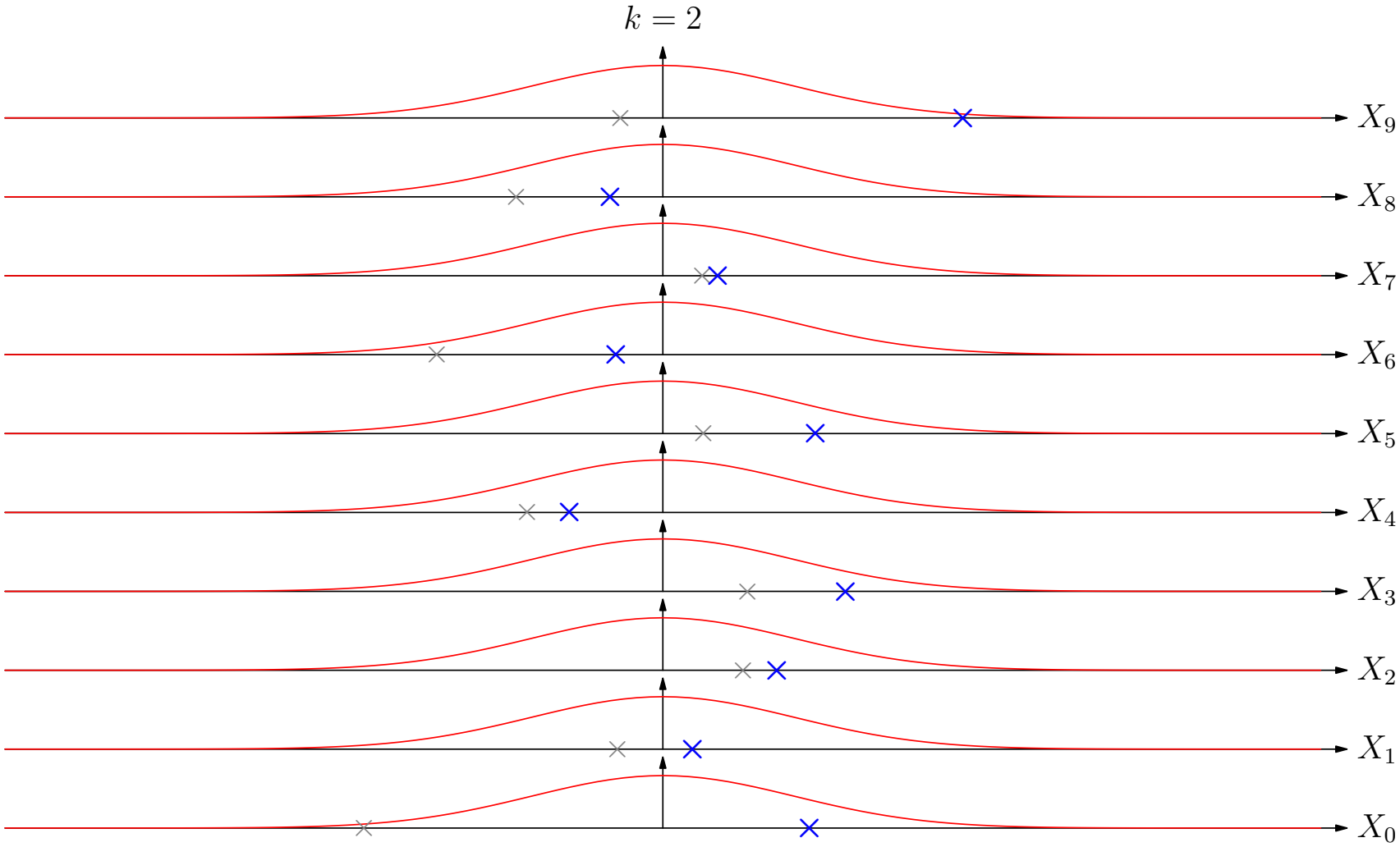

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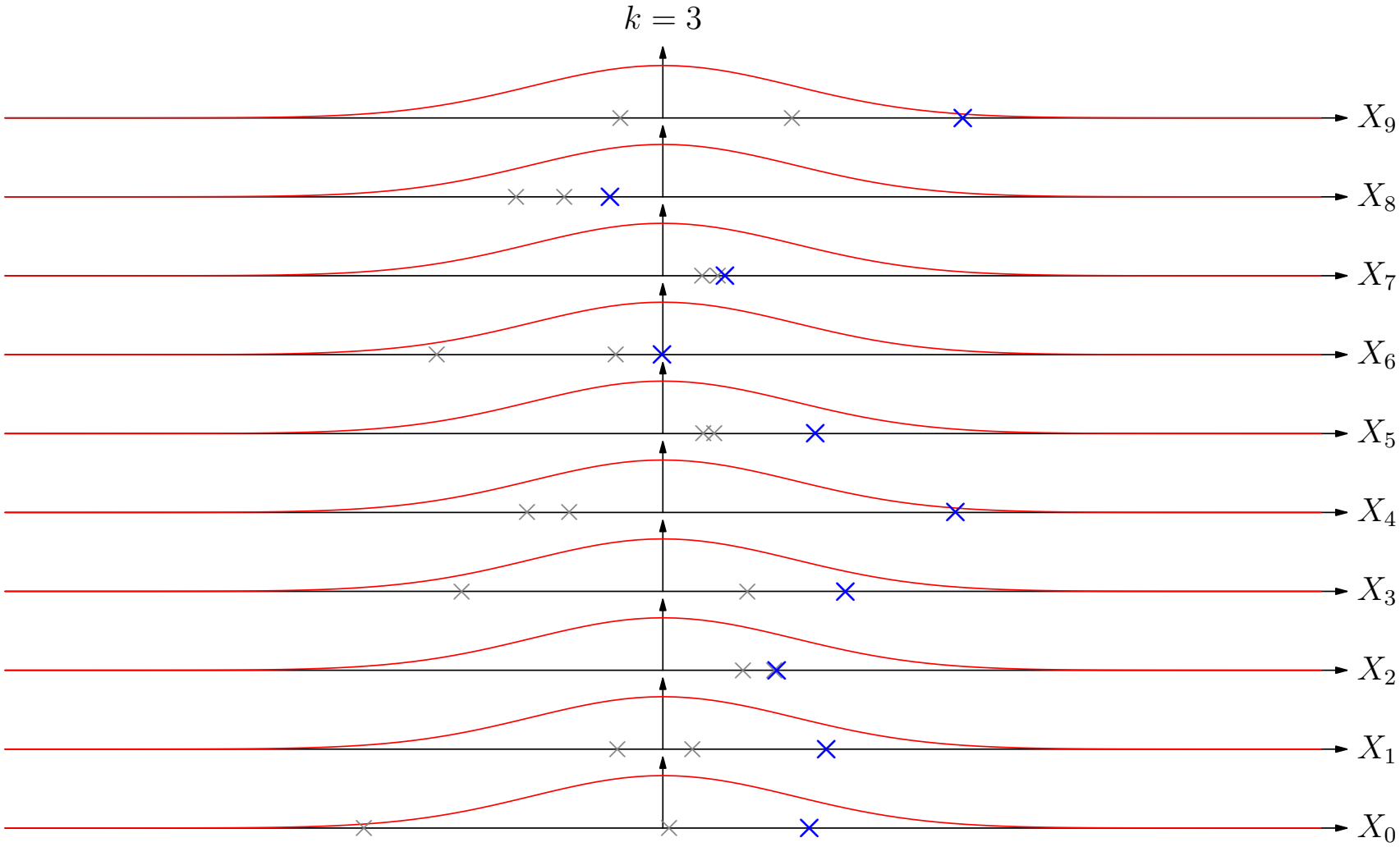

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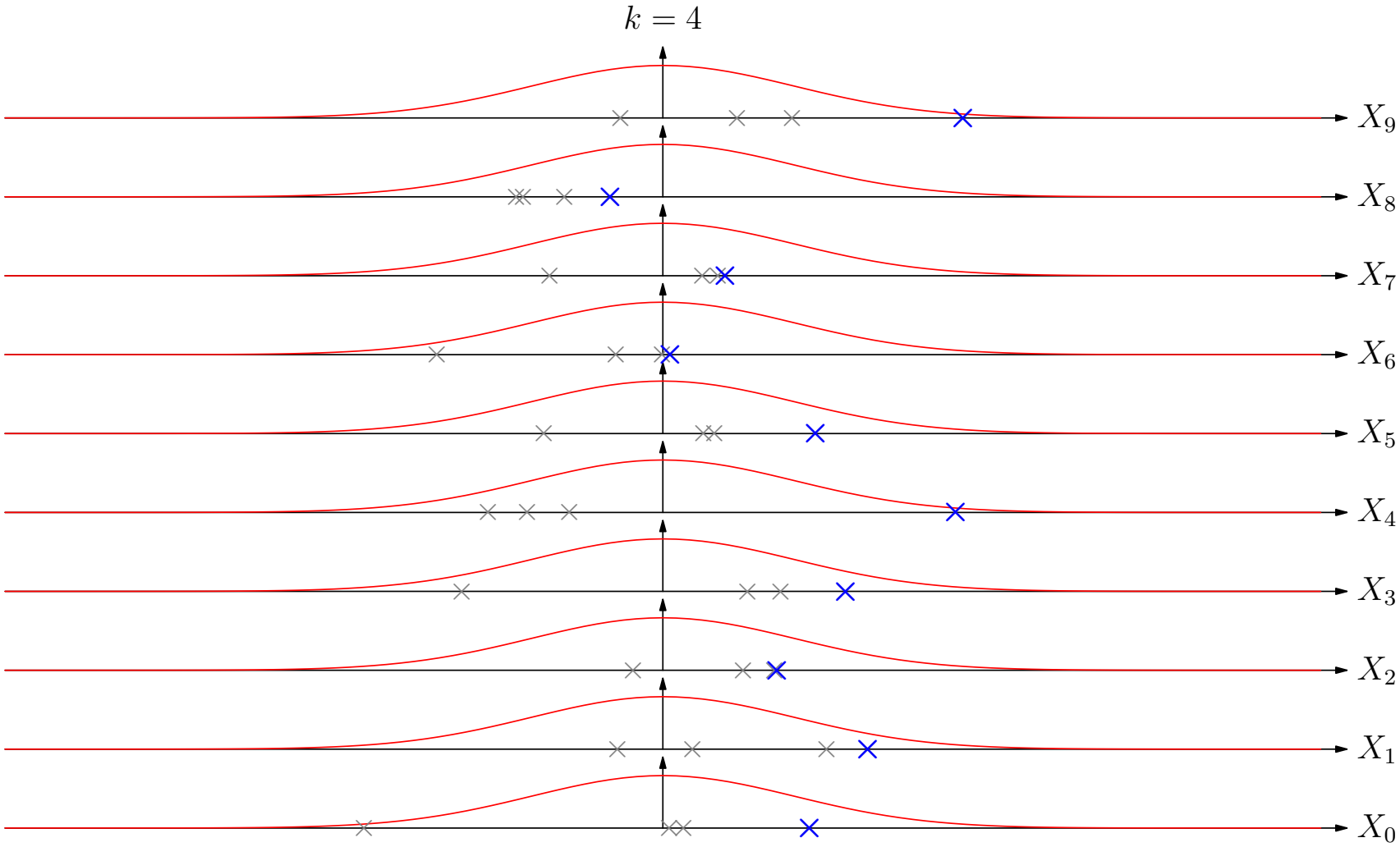

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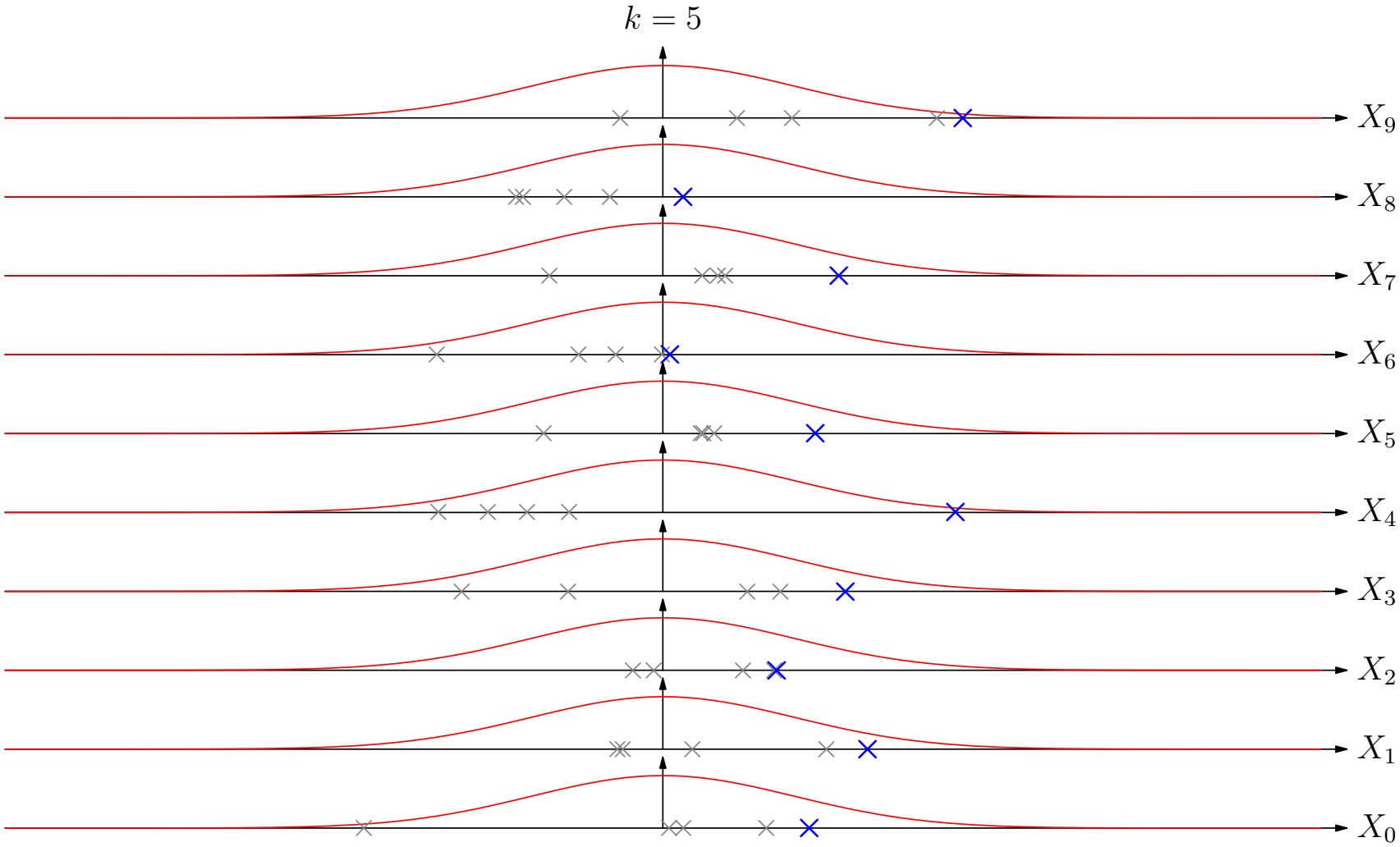

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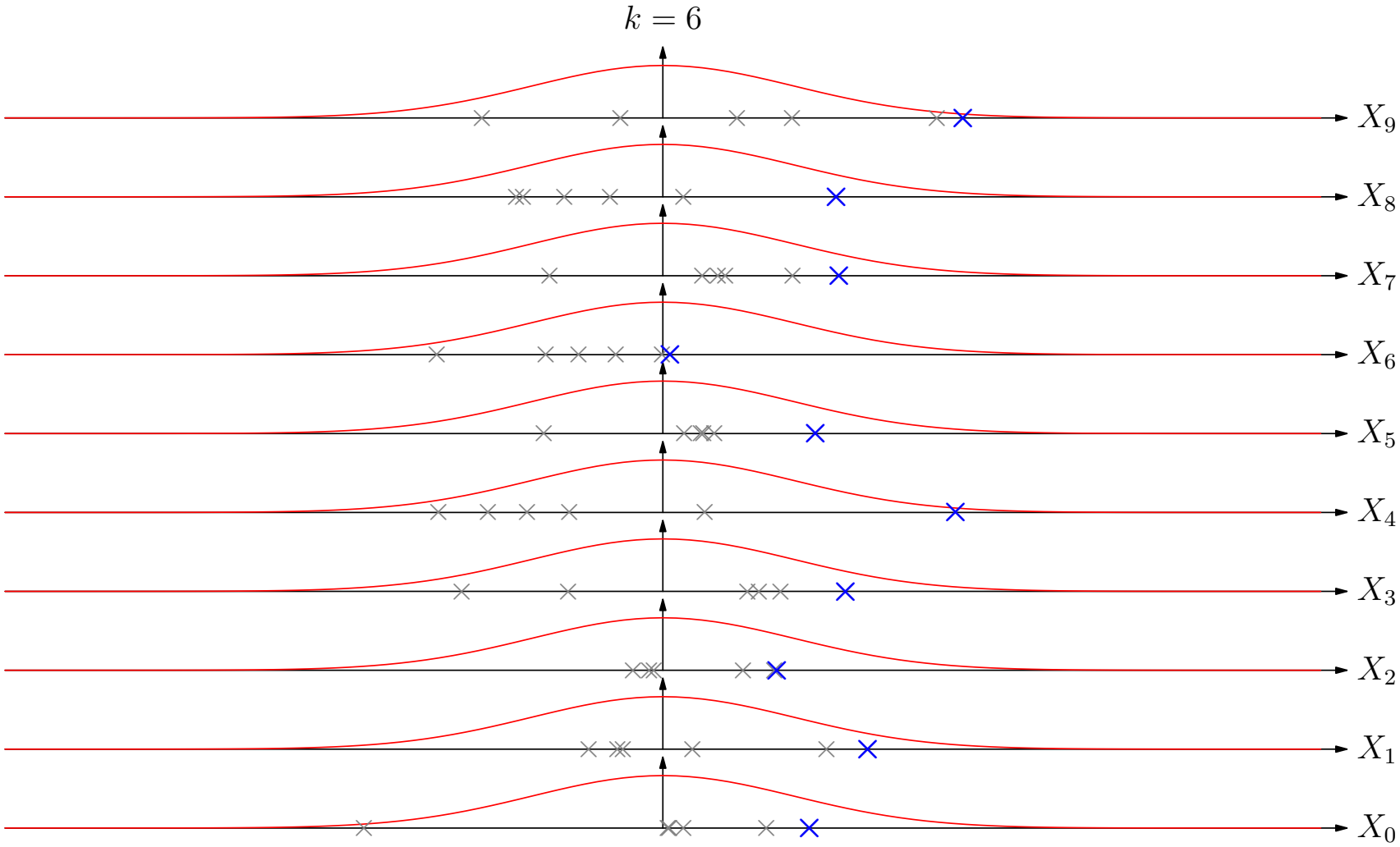

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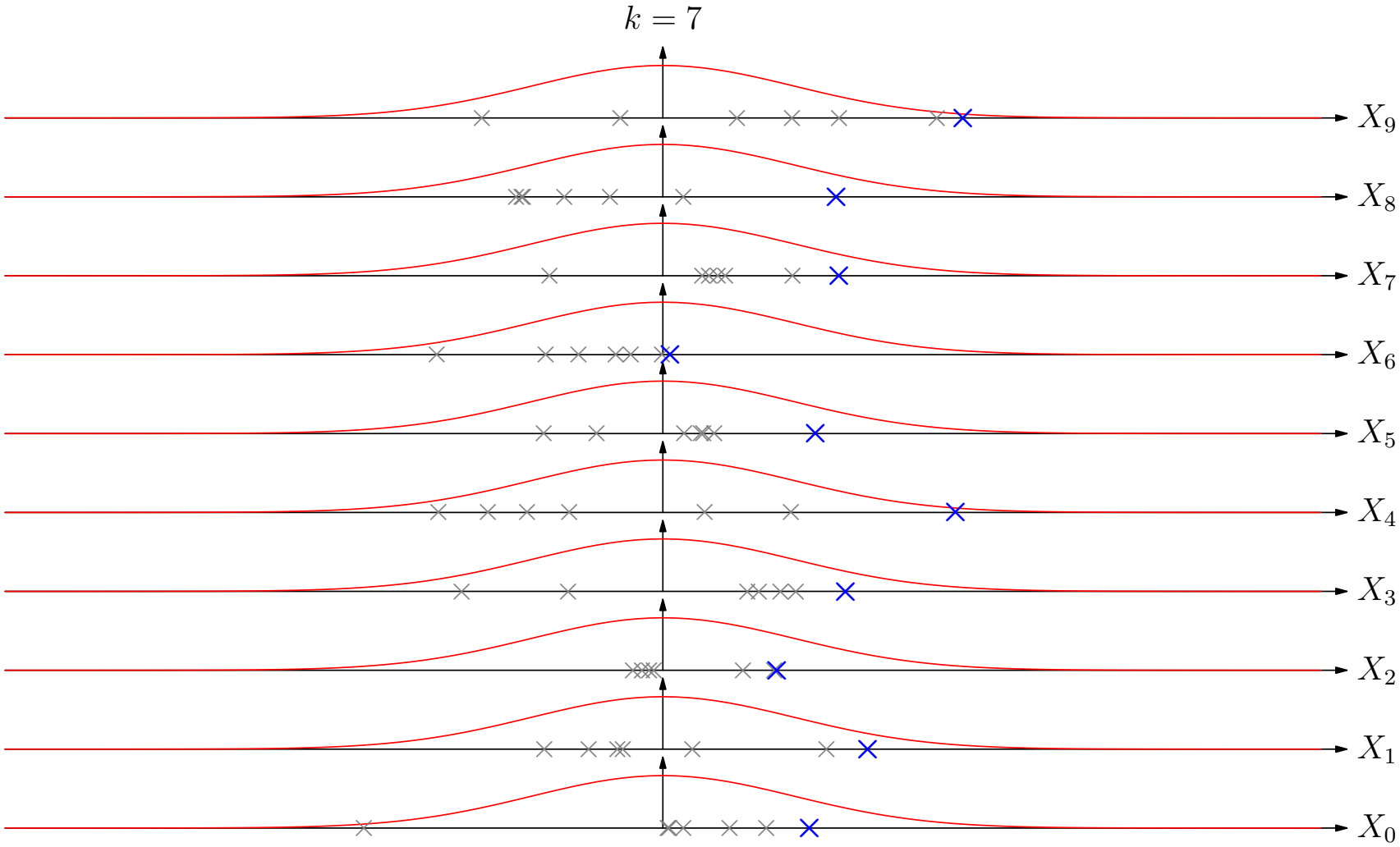

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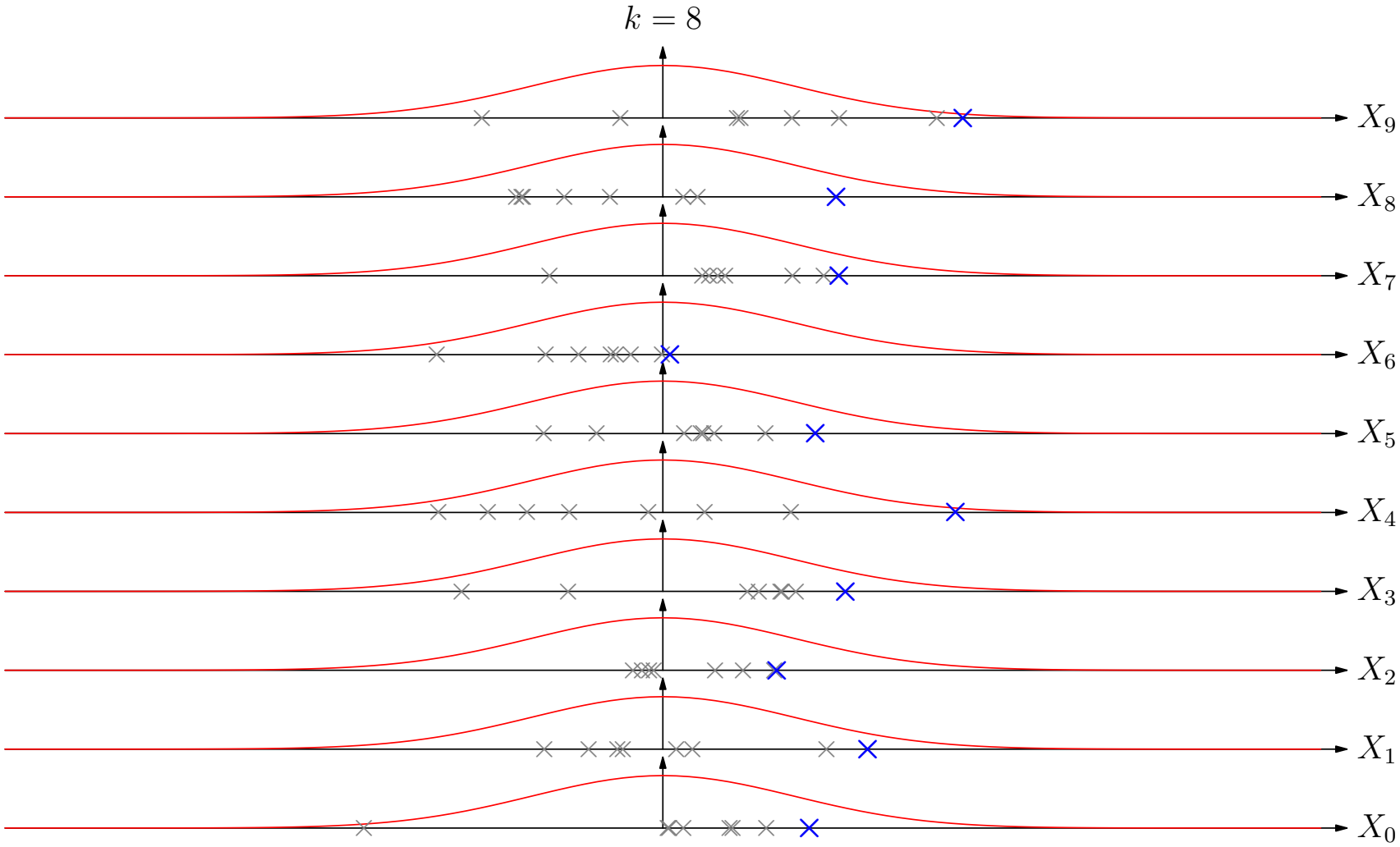

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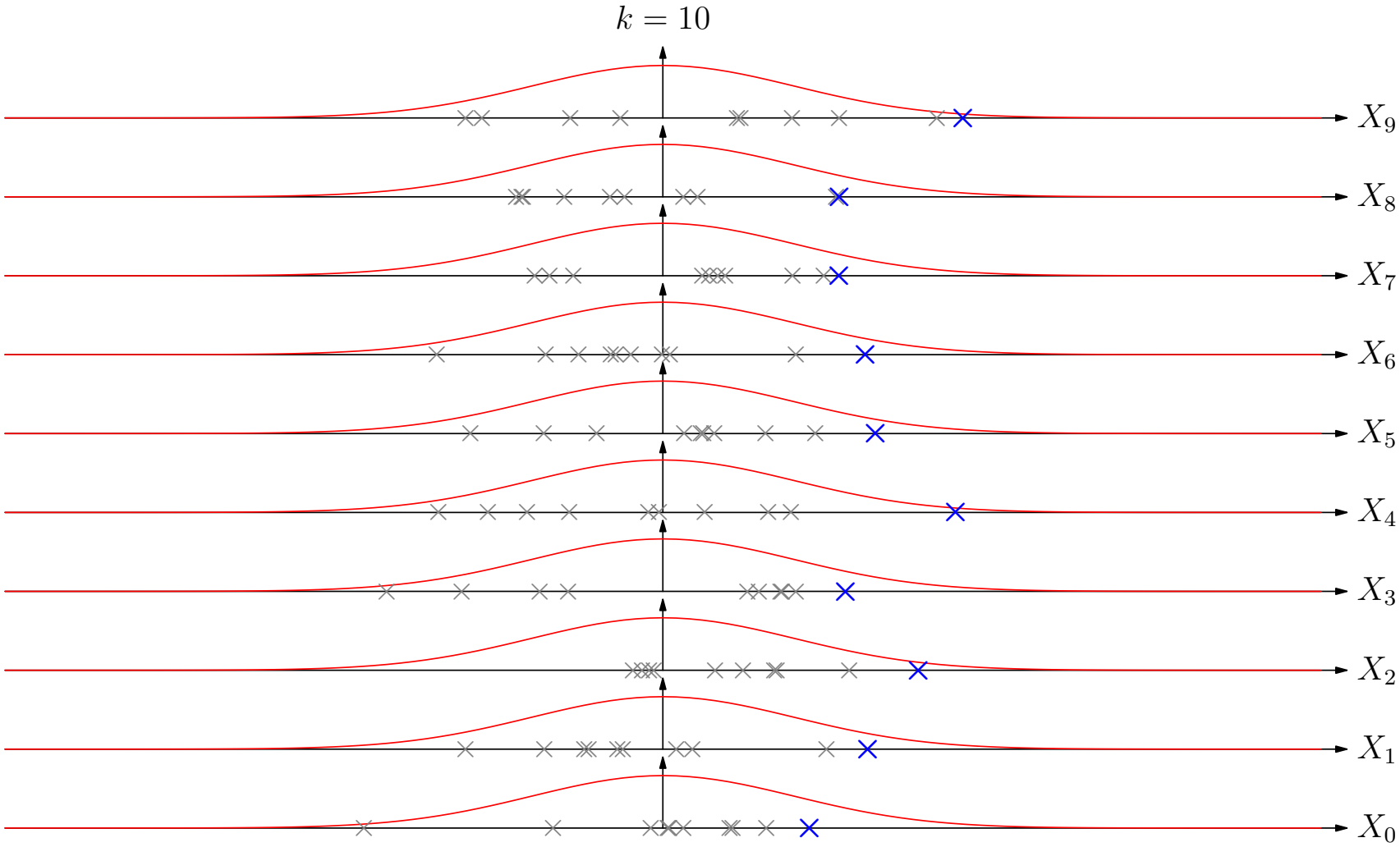

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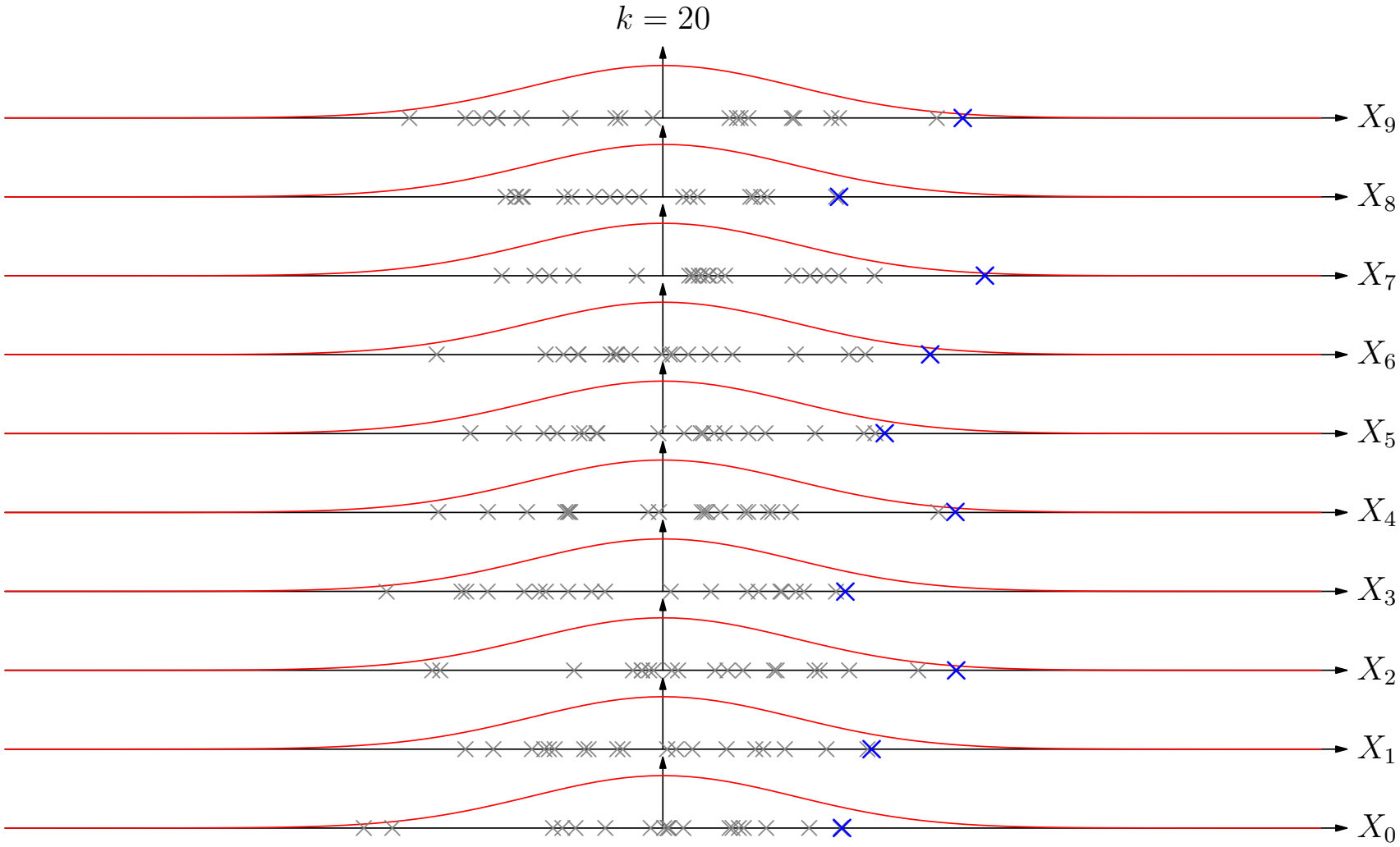

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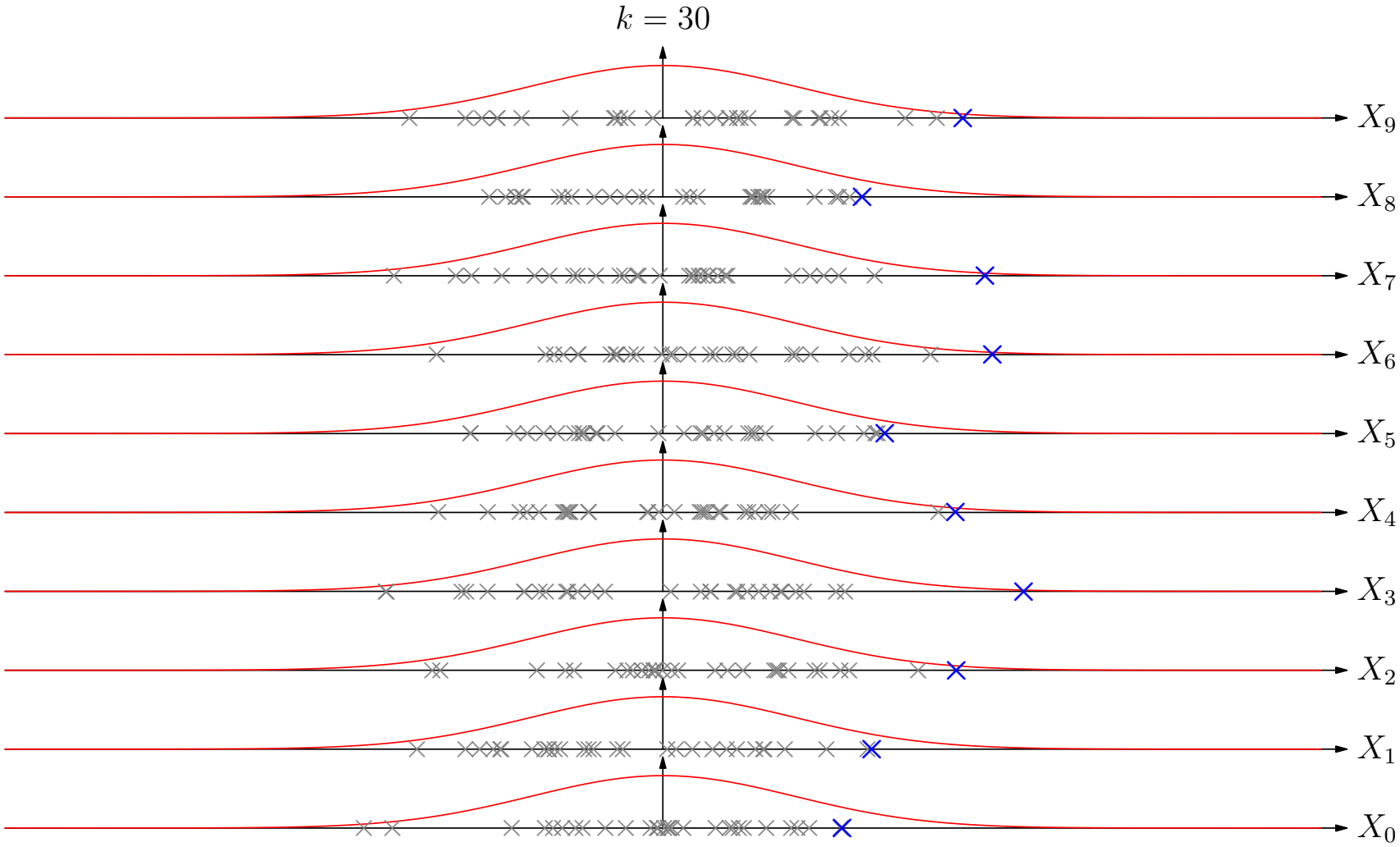

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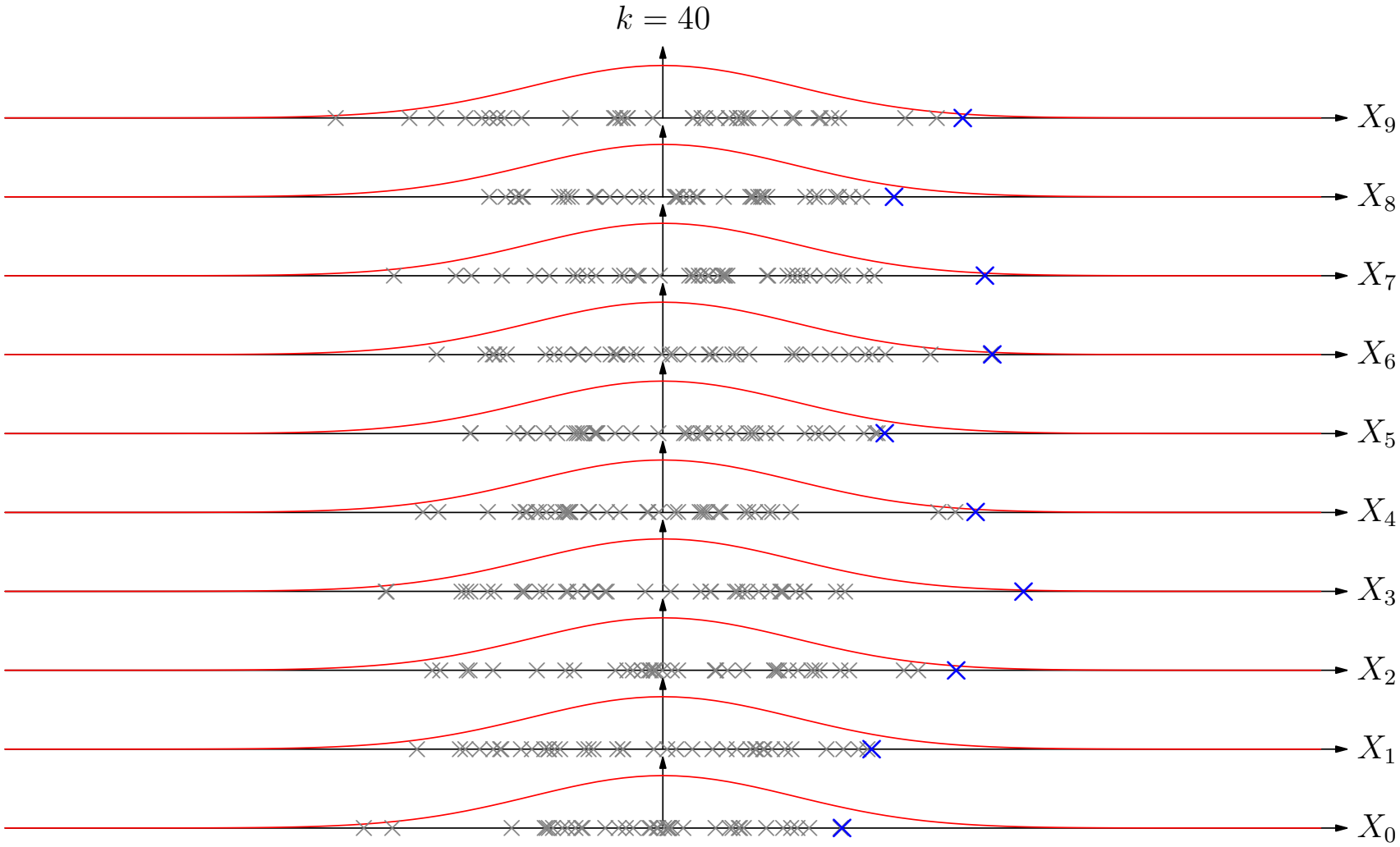

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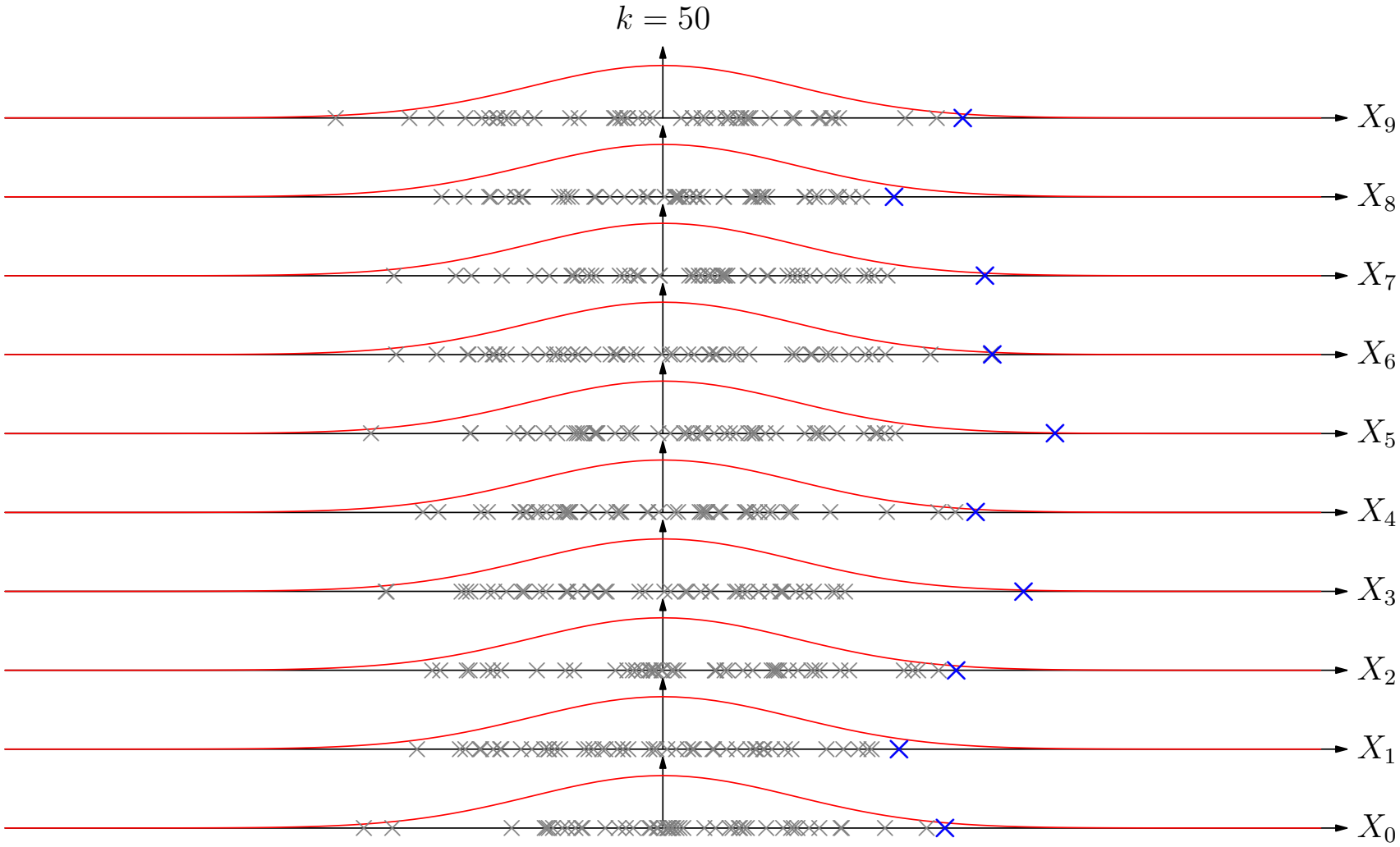

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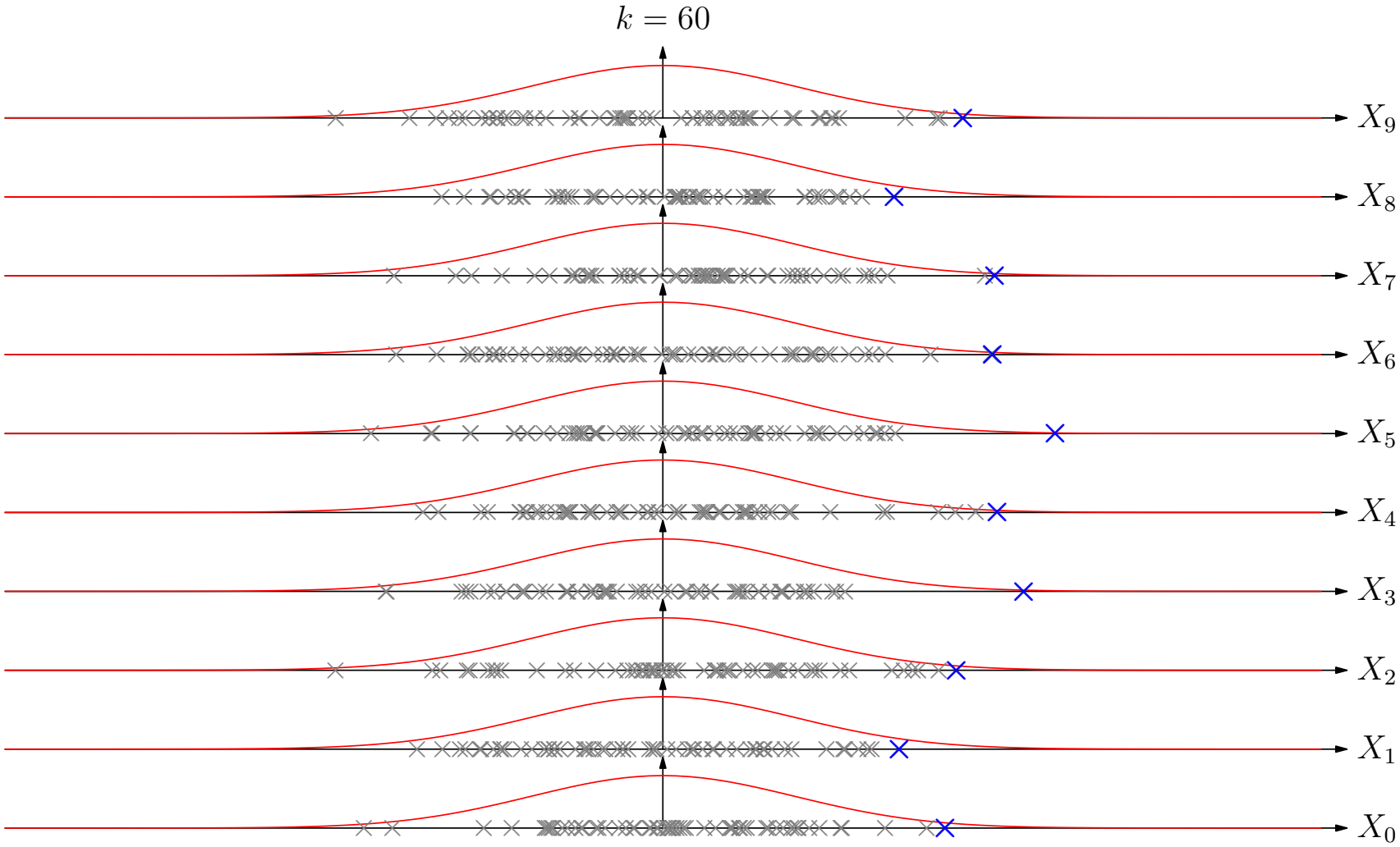

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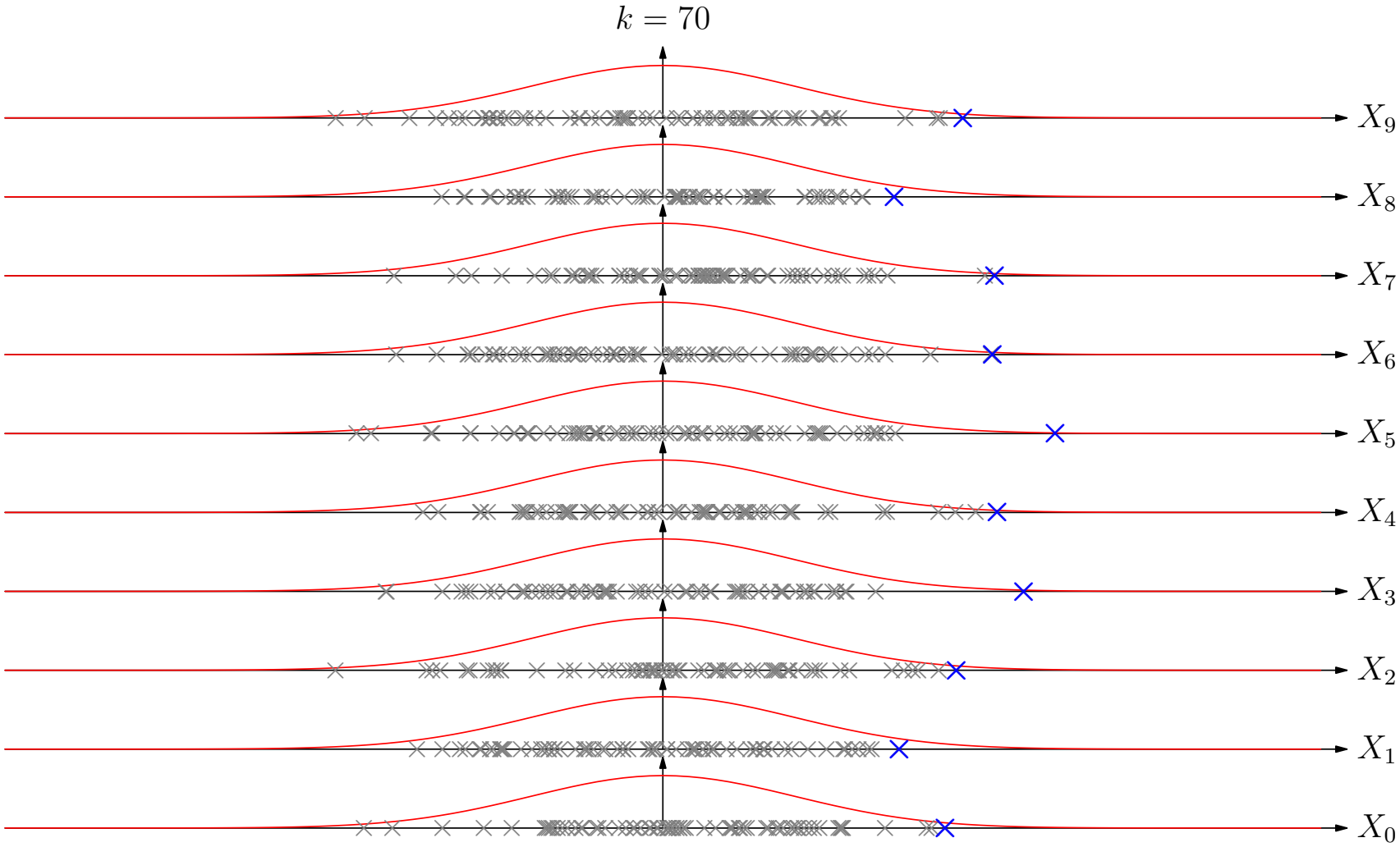

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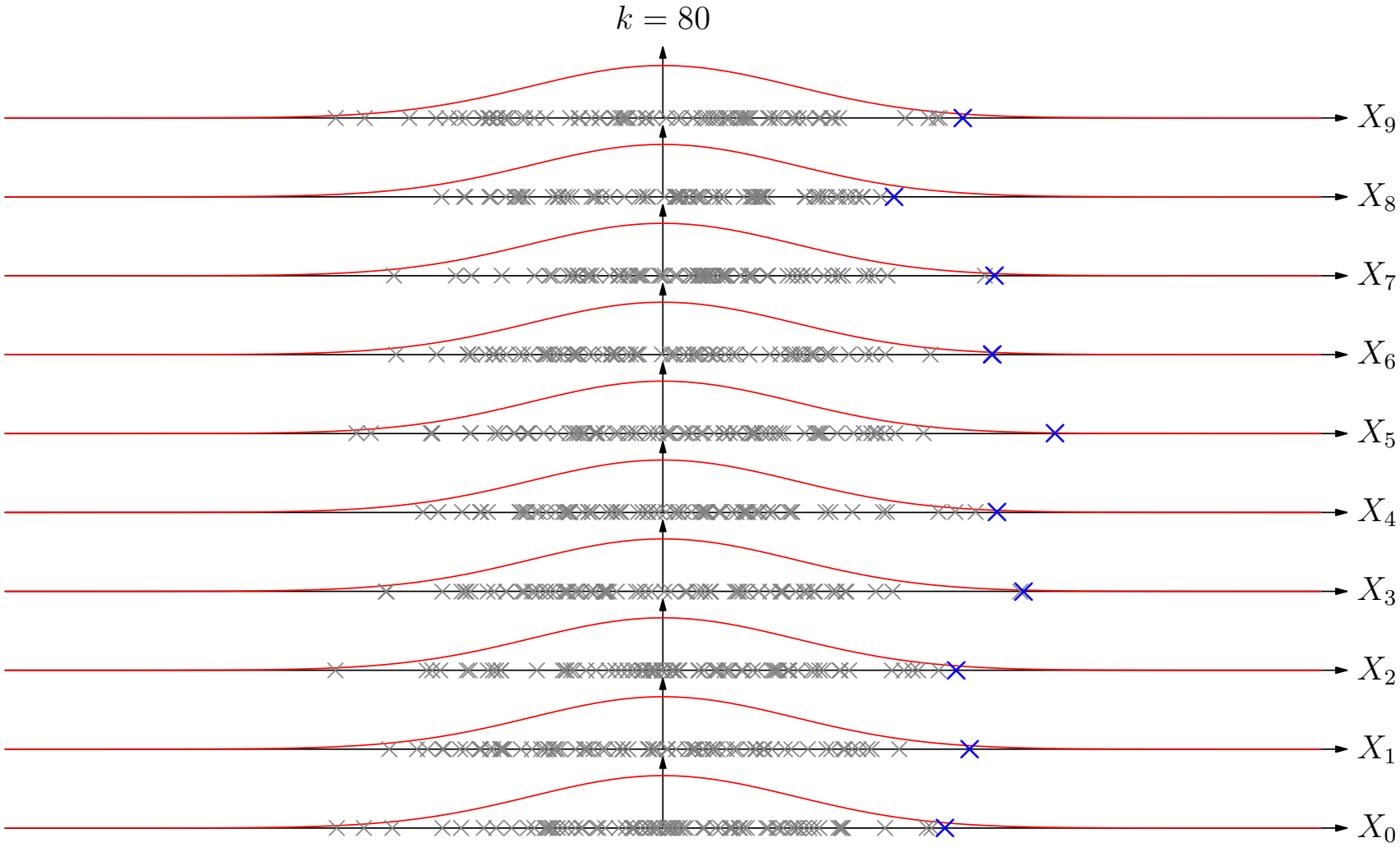

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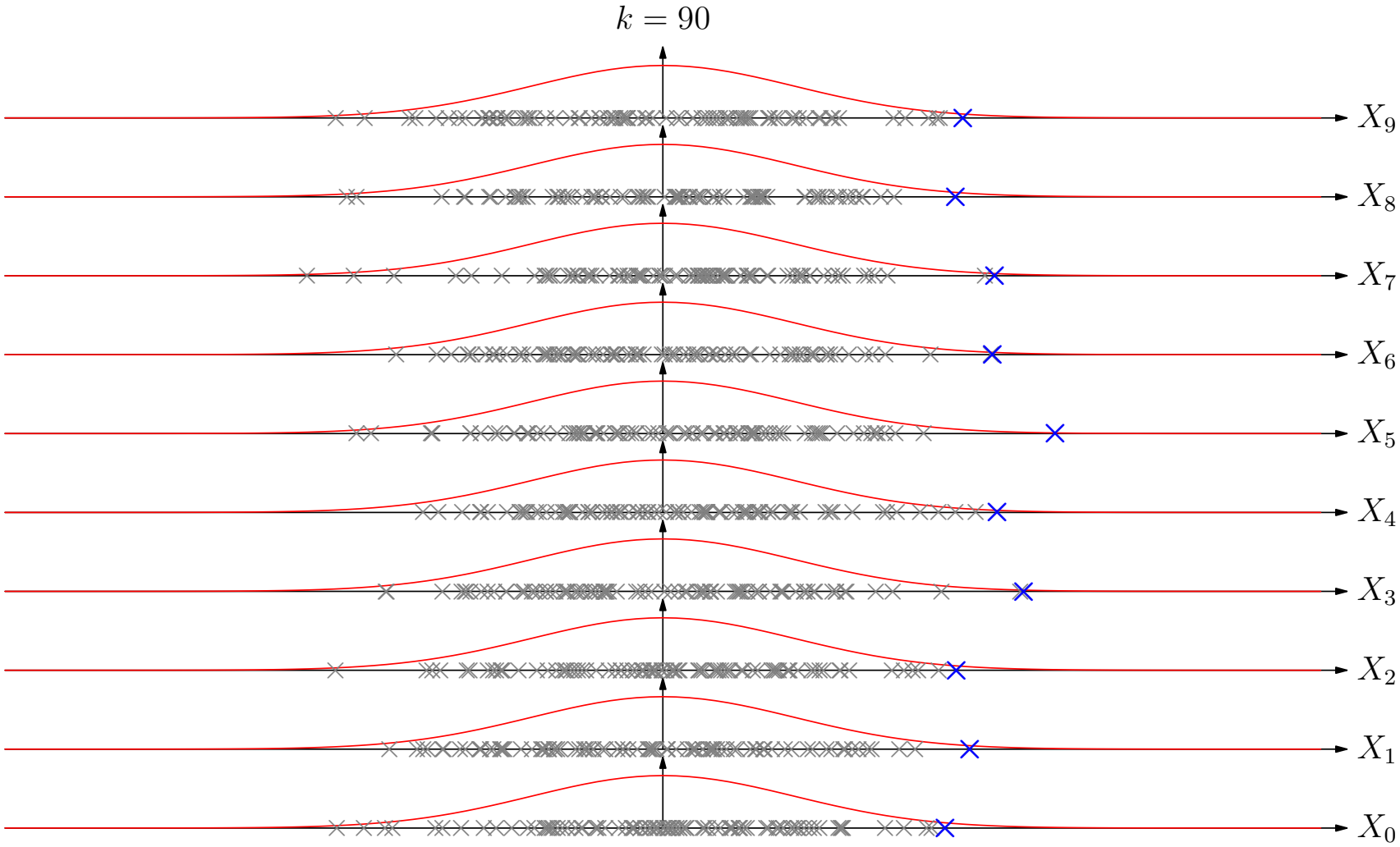

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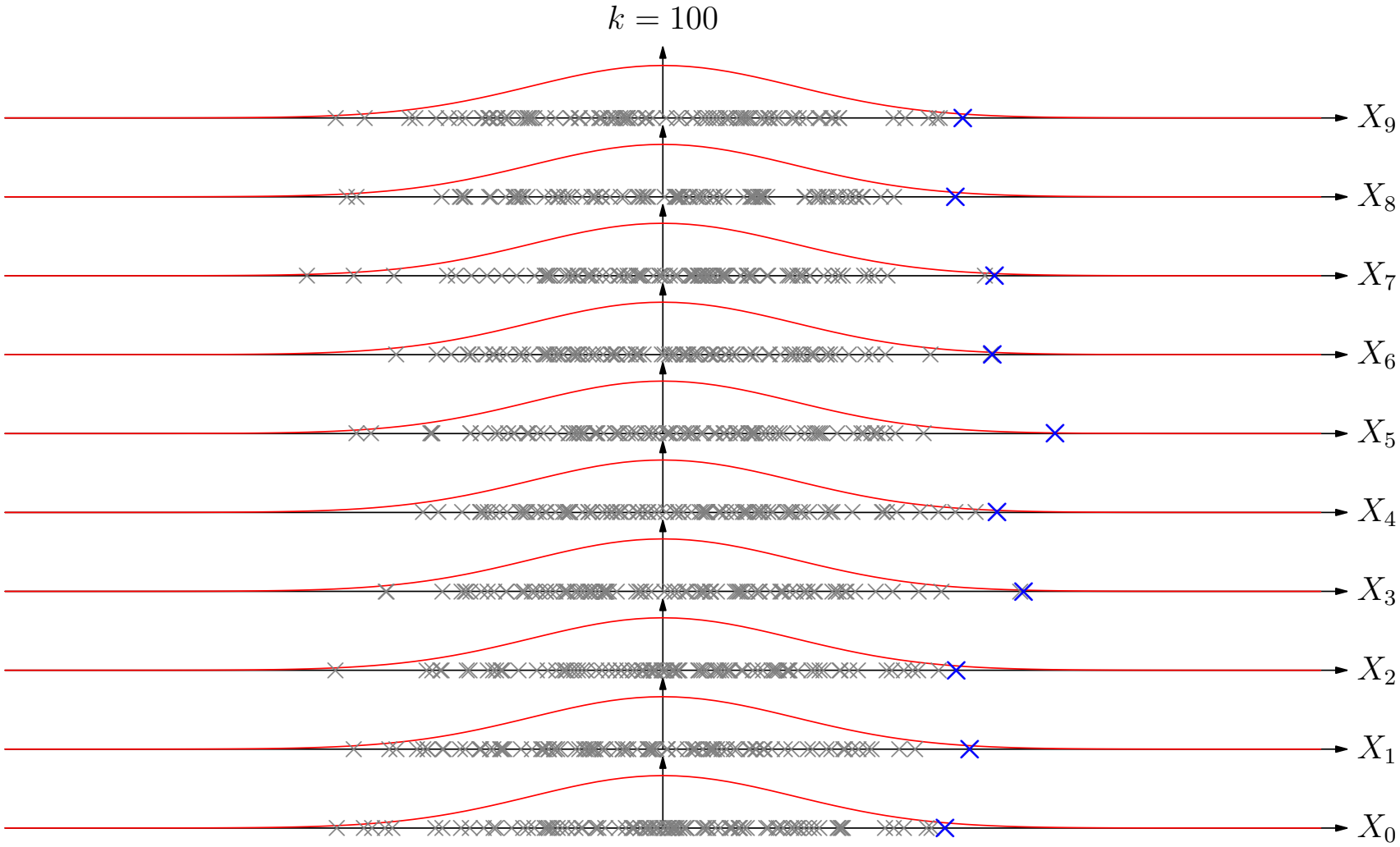

Extreme Value Distribution

- The distribution of extreme values (best of k) is known as the extreme value distribution

- Similar to the central limit theorem (which states that the sum of random variables will be normally distributed), it turns out that asymptotically ($k \to \infty$) the extreme value of random deviates converges to a small number of extreme value distributions

- The most common is the Gumbel EVD given by

$$f(z) \approx \frac{1}{\beta} \mathrm{e}^{-\frac{x-\mu}{\beta} - e^{-\frac{x-\mu}{\beta}}}$$

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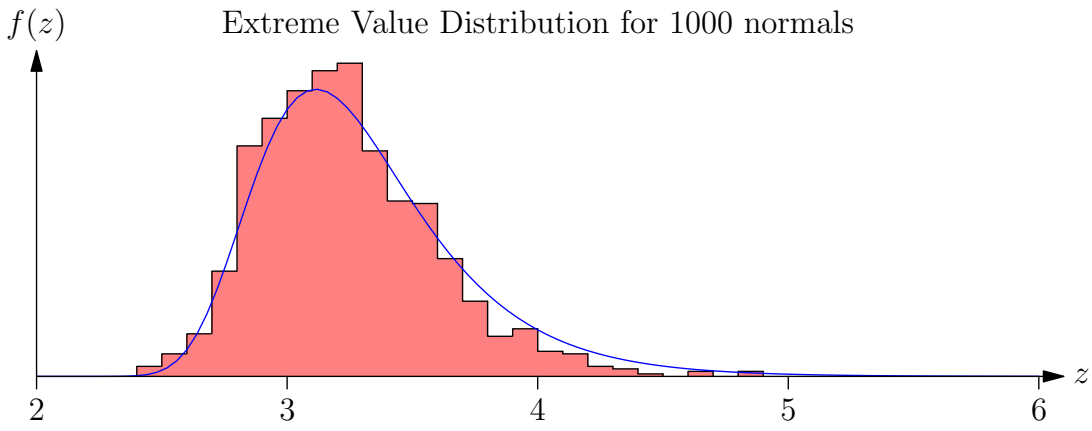
Using EVD

- Extreme Value Distributions become more accurate when we do a large number of trials



- In matching the number of matches are exponential in the string length so we are in a situation where EVDs are quite accurate
- The probability of observing a match with score at least Z is

$$\mathbb{P}\left(z \geq Z\right) = \int_{Z}^{\infty} f(z)\,\mathrm{d}z = 1 - \exp\left(-e^{-\frac{x-\mu}{\beta}}\right)$$

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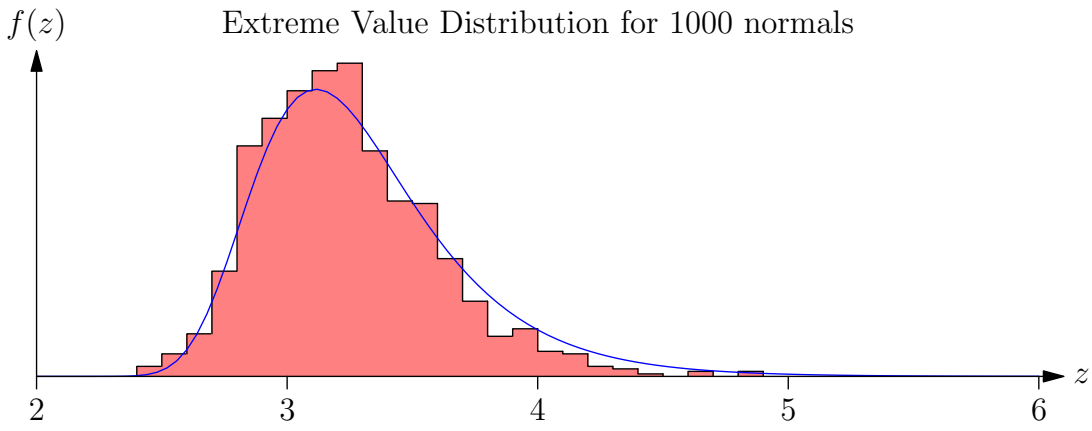

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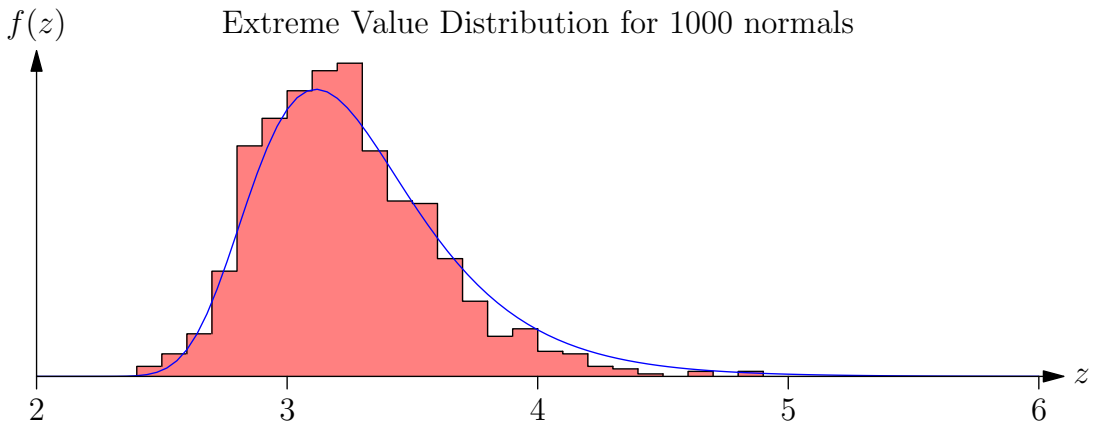

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